

DataAnalysisUpdate

September 18, 2024

1 Data Dictionary

Column Name	Description
SEQN	unique identifier
BMDSTATS	Body Measures Component Status Code
BMXBMI	BMI
BMXLEG	Upper Leg Length (cm)
BMXWAIST	Waist Circumference (cm)
BMDBMIC	BMI Category
RIDSTATR	Interview/Examination status
RIAGENDR	Gender of the participant
RIDAGEYR	Age in years at screening
RIDRETH1	Race/Hispanic origin
RIDRETH3	Recode of reported race and Hispanic origin information, with Non-Hispanic Asian Category
RIDEXMON	Six month time period when the examination was performed - two categories: November 1 through April 30, May 1 through October 31
LUXCAPM	Liver Ultrasound Transient Elastography test - Median CAP, decibels per meter (dB/m)
LUAXSTAT	Elastography exam status
LBXIN	Insulin (uU/mL)
ALQ151	Ever have 4/5 or more drinks every day?
BPQ080	Doctor told you - high cholesterol level
BPQ100D	Blood Pressure & Cholesterol Now taking prescribed medicine
CDQ010	Shortness of breath on stairs/walking up a slight hill
DIQ070	Take diabetic pills to lower blood sugar
MCQ053	Taking treatment for anemia/past 3 mos
OCQ180	Hours worked last week in total all jobs
LBXTC	Total Cholesterol (mg/dL)
interview__day	Added value for datetime

2 Introduction to DataFrame

2.1 Import Data

```
[52]: import pandas as pd
import seaborn as sns
import numpy as np

nhane = pd.read_csv('my_nhane.csv')

days = pd.DataFrame({'date':pd.date_range('2017-01-01','2019-12-31',freq='D')})
date = days.sample(n=nhane.shape[0],replace=True)
date = date.reset_index()
date.drop(columns='index',axis=1,inplace=True)
nhane['interview_day'] = date
```

2.2 Data Frame Structure

```
[53]: nhane[['SEQN','BMXBMI','RIDSTATR','RIDRETH1','BMXHT']].head()
```

```
[53]:
```

	SEQN	BMXBMI	RIDSTATR \
0	109263	NaN	Both interviewed and MEC examined
1	109264	17.6	Both interviewed and MEC examined
2	109265	15.0	Both interviewed and MEC examined
3	109266	37.8	Both interviewed and MEC examined
4	109267	NaN	Interviewed only

		RIDRETH1	BMXHT
0	Other Race - Including Multi-Racial		NaN
1		Mexican American	154.7
2		Non-Hispanic White	89.3
3	Other Race - Including Multi-Racial		160.2
4		Other Hispanic	NaN

```
[ ]:
```

2.3 Series Structure

```
[54]: nhane['BMXBMI']
```

```
[54]:
```

0	NaN
1	17.6
2	15.0
3	37.8
4	NaN
...	
15577	29.5

```

15578    15.9
15579    32.0
15580    39.4
15581    14.9
Name: BMXBMI, Length: 15582, dtype: float64

```

3 Read Dataset

3.1 Read csv and read excel

- https://pandas.pydata.org/pandas-docs/stable/reference/api/pandas.read_csv.html
- https://pandas.pydata.org/pandas-docs/stable/reference/api/pandas.read_excel.html

```
[55]: pd.read_csv('test1.csv',delimiter=';',skiprows=3)
```

```
[55]:
```

	Name	Age	Location	Company
0	Nik	34	Toronto	datagy
1	Kate	33	New York City	Apple

```
[56]: pd.read_excel('test2.xlsx',sheet_name='employees',skiprows = 2)
```

```
[56]:
```

	Unnamed: 0	EmpID	EmpName	EmpRole	Unnamed: 4
0	0	1	Pankaj	CEO	NaN
1	1	2	David	Lee	Editor
2	2	3	Lisa	Ray	Author

4 Data Frame Indexing – loc iloc

4.1 loc (-label)

4.1.1 1.Selecting a Single Row

```
[57]: # 1. Selecting a Single Row
nhane.loc[2]
```

```
[57]:
```

Unnamed: 0	3
SEQN	109265
BPQ080	NaN
BPQ100D	NaN
BMDSTATS	Complete data for age group
BMXHT	89.3
BMXBMI	15.0
BMDBMIC	Normal weight
BMXLEG	NaN
BMXWAIST	41.2
RIDSTATR	Both interviewed and MEC examined
RIAGENDR	Male
RIDAGEYR	2

```

RIDRETH1                Non-Hispanic White
RIDRETH3                Non-Hispanic White
RIDEXMON                May 1 through October 31
DIQ070                  NaN
CDQ010                  NaN
ALQ151                  NaN
OCQ180                  NaN
LUAXSTAT                NaN
LUXCAPM                 NaN
LBXIN                   NaN
MCQ053                  No
LBXTC                   NaN
interview_day            2018-10-08 00:00:00
Name: 2, dtype: object

```

Question: - what is the type of this `nhane.loc[2]` - what is the index of `nhane.loc[2]`

4.1.2 2. Selecting a Single column

```
[58]: nhane.loc[:, 'BMXBMI']
```

```

[58]: 0      NaN
      1      17.6
      2      15.0
      3      37.8
      4      NaN
      ...
      15577  29.5
      15578  15.9
      15579  32.0
      15580  39.4
      15581  14.9
      Name: BMXBMI, Length: 15582, dtype: float64

```

Selecting a Single column using column name directly

```
[59]: nhane['BMXBMI']
```

```

[59]: 0      NaN
      1      17.6
      2      15.0
      3      37.8
      4      NaN
      ...
      15577  29.5
      15578  15.9
      15579  32.0
      15580  39.4

```

```
15581    14.9
Name: BMXBMI, Length: 15582, dtype: float64
```

Selecting multiple columns using column name directly

```
[60]: nhane[['SEQN', 'BMXBMI']]
```

```
[60]:      SEQN  BMXBMI
0      109263      NaN
1      109264      17.6
2      109265      15.0
3      109266      37.8
4      109267      NaN
...
15577  115983      29.5
15578  115984      15.9
15579  115985      32.0
15580  115986      39.4
15581  115987      14.9

[15582 rows x 2 columns]
```

4.1.3 3. Selecting a multiple rows

Slicing

```
[61]: nhane.loc[0:3] # watch out the index number
```

```
[61]: Unnamed: 0      SEQN BPQ080 BPQ100D      BMDSTATS  BMXHT  \
0          1  109263      NaN      NaN  No body measures exam data      NaN
1          2  109264      NaN      NaN  Complete data for age group  154.7
2          3  109265      NaN      NaN  Complete data for age group   89.3
3          4  109266      Yes      NaN  Complete data for age group  160.2

      BMXBMI      BMDBMIC  BMXLEG  BMXWAIST  ...  DIQ070  CDQ010  ALQ151  OCQ180  \
0      NaN      NaN      NaN      NaN  ...      NaN      NaN      NaN      NaN
1  17.6  Normal weight    36.3    63.8  ...      NaN      NaN      NaN      NaN
2  15.0  Normal weight      NaN    41.2  ...      NaN      NaN      NaN      NaN
3  37.8      NaN    40.8   117.9  ...      No      NaN      No    40.0

      LUAXSTAT  LUXCAPM  LBXIN  MCQ053  LBXTC  interview_day
0      NaN      NaN      NaN      No      NaN    2017-10-11
1  Ineligible      NaN    6.05      No  166.0    2019-01-02
2      NaN      NaN      NaN      No      NaN    2018-10-08
3  Complete    277.0      NaN      No  195.0    2019-04-10

[4 rows x 26 columns]
```

Use label index

```
[62]: nhane.loc[[1,3,5]]
```

```
[62]:      Unnamed: 0      SEQN BPQ080 BPQ100D      BMDSTATS  BMXHT  \
1          2  109264      NaN      NaN  Complete data for age group  154.7
3          4  109266      Yes      NaN  Complete data for age group  160.2
5          6  109268      No      NaN                                NaN    NaN

      BMXBMI      BMDBMIC  BMXLEG  BMXWAIST  ... DIQ070 CDQ010  ALQ151  OCQ180  \
1    17.6  Normal weight    36.3    63.8  ...    NaN    NaN    NaN    NaN
3    37.8              NaN    40.8   117.9  ...    No    NaN    No   40.0
5     NaN              NaN    NaN     NaN  ...    NaN    NaN    NaN    NaN

      LUAXSTAT  LUXCAPM  LBXIN  MCQ053  LBXTC  interview_day
1  Ineligible      NaN  6.05      No  166.0    2019-01-02
3   Complete    277.0  NaN      No  195.0    2019-04-10
5        NaN      NaN  NaN      No   NaN    2019-04-07

[3 rows x 26 columns]
```

4.1.4 4. Selecting Multiple Rows and Multiple Columns

```
[63]: nhane.loc[[1,5,7], ['SEQN', 'BMXBMI', 'RIDSTATR', 'RIDRETH1']]
```

```
[63]:      SEQN  BMXBMI      RIDSTATR      RIDRETH1
1  109264    17.6  Both interviewed and MEC examined  Mexican American
5  109268     NaN              Interviewed only  Non-Hispanic White
7  109270    30.9  Both interviewed and MEC examined  Non-Hispanic Black
```

```
[ ]:
```

4.1.5 Practice

Column Name	Description
SEQN	unique identifier
BMDSTATS	Body Measures Component Status Code
BMXBMI	BMI

```
loc 10 80 , 90 ,
```

```
[64]: lst = list(range(10,81))+[90]
nhane.loc[lst, ['SEQN', 'BMDSTATS', 'BMXBMI']]
```

```
[64]:      SEQN      BMDSTATS  BMXBMI
10  109273      Complete data for age group    21.9
11  109274      Complete data for age group    30.2
12  109275  Partial: Only height and weight obtained    14.4
```

13	109276	Complete data for age group	NaN
14	109277	Complete data for age group	18.6
..	...	...	...
77	109340	Complete data for age group	46.2
78	109341	Complete data for age group	15.8
79	109342	Complete data for age group	35.5
80	109343	NaN	NaN
90	109353	Complete data for age group	21.4

[72 rows x 3 columns]

pd.read_csv pd.read_excel test1.csv ** 1.xlsx

[]:

loc 25 5

[65]: nhane.loc[25, 'BMXHT'] = 109# 25 5

[66]: nhane.loc[25, 'BMXHT']

[66]: 109.0

4.2 iloc []

4.2.1 1. Selecting a Single Row

[67]: nhane.iloc[3]

[67]: Unnamed: 0 4
 SEQN 109266
 BPQ080 Yes
 BPQ100D NaN
 BMDSTATS Complete data for age group
 BMXHT 160.2
 BMXBMI 37.8
 BMDBMIC NaN
 BMXLEG 40.8
 BMXWAIST 117.9
 RIDSTATR Both interviewed and MEC examined
 RIAGENDR Female
 RIDAGEYR 29
 RIDRETH1 Other Race - Including Multi-Racial
 RIDRETH3 Non-Hispanic Asian
 RIDEXMON May 1 through October 31
 DIQ070 No
 CDQ010 NaN
 ALQ151 No
 OCQ180 40.0

```

LUAXSTAT                                Complete
LUXCAPM                                277.0
LBXIN                                  NaN
MCQ053                                No
LBXTC                                  195.0
interview_day                          2019-04-10 00:00:00
Name: 3, dtype: object

```

4.2.2 2. Selecting a Single column

```
[68]: nhane.iloc[:, 5]
```

```

[68]: 0      NaN
      1    154.7
      2     89.3
      3    160.2
      4      NaN
      ...
    15577  160.8
    15578  115.9
    15579  173.2
    15580  164.8
    15581  116.6
      Name: BMXHT, Length: 15582, dtype: float64

```

4.2.3 3. Selecting a multiple rows

Slicing

```
[69]: nhane.iloc[1:3]
```

```

[69]: Unnamed: 0    SEQN BPQ080 BPQ100D          BMDSTATS  BMXHT  \
      1          2  109264    NaN    NaN  Complete data for age group  154.7
      2          3  109265    NaN    NaN  Complete data for age group   89.3

      BMXBMI      BMDBMIC  BMXLEG  BMXWAIST  ...  DIQ070  CDQ010  ALQ151  OCQ180  \
      1   17.6  Normal weight    36.3    63.8  ...    NaN    NaN    NaN    NaN
      2   15.0  Normal weight    NaN    41.2  ...    NaN    NaN    NaN    NaN

      LUAXSTAT  LUXCAPM  LBXIN  MCQ053  LBXTC  interview_day
      1  Ineligible    NaN   6.05    No  166.0    2019-01-02
      2      NaN     NaN   NaN    No   NaN    2018-10-08

      [2 rows x 26 columns]

```

using label index

```
[70]: nhane.iloc[[1,3,5]]
```

```
[70]: Unnamed: 0      SEQN BPQ080 BPQ100D      BMDSTATS  BMXHT  \
1          2  109264      NaN      NaN  Complete data for age group  154.7
3          4  109266      Yes      NaN  Complete data for age group  160.2
5          6  109268      No      NaN      NaN      NaN

      BMXBMI      BMDBMIC  BMXLEG  BMXWAIST  ... DIQ070 CDQ010  ALQ151  OCQ180  \
1    17.6  Normal weight    36.3    63.8  ...   NaN   NaN    NaN   NaN
3    37.8      NaN    40.8   117.9  ...   No   NaN    No   40.0
5     NaN      NaN    NaN     NaN  ...   NaN   NaN    NaN   NaN

      LUAXSTAT  LUXCAPM  LBXIN  MCQ053  LBXTC  interview_day
1  Ineligible      NaN  6.05     No  166.0    2019-01-02
3   Complete    277.0  NaN     No  195.0    2019-04-10
5      NaN      NaN  NaN     No   NaN    2019-04-07

[3 rows x 26 columns]
```

4.2.4 4. Selecting Multiple Rows and Multiple Columns

```
[71]: nhane.iloc[1:3, [1, 5, 9, 12]]
```

```
[71]:      SEQN  BMXHT  BMXWAIST  RIDAGEYR
1  109264  154.7     63.8      13
2  109265   89.3     41.2       2
```

4.2.5 Practice

Column Name	Description
SEQN	unique identifier
BMDSTATS	Body Measures Component Status Code
BMXBMI	BMI

```
iloc 10 80 , 90 ,
```

```
[72]: list(range(10,81))+[90]
nhane.iloc[list(range(10,81))+[90], [1,4,6]]
```

```
[72]:      SEQN      BMDSTATS  BMXBMI
10  109273      Complete data for age group  21.9
11  109274      Complete data for age group  30.2
12  109275  Partial: Only height and weight obtained  14.4
13  109276      Complete data for age group   NaN
14  109277      Complete data for age group  18.6
..   ...
77  109340      Complete data for age group  46.2
78  109341      Complete data for age group  15.8
79  109342      Complete data for age group  35.5
```

```

80 109343          NaN      NaN
90 109353      Complete data for age group      21.4

```

[72 rows x 3 columns]

```
iloc    25  5
```

```
[73]: nhane.iloc[24, 5]
```

```
[73]: 145.1
```

4.3 Comparison between loc and iloc method

```
[74]: nhane.iloc[1:4, 1:4]
```

```
[74]:      SEQN  BPQ080  BPQ100D
1  109264      NaN      NaN
2  109265      NaN      NaN
3  109266      Yes      NaN

```

```
[75]: nhane.loc[1:4, 'SEQN' : 'BPQ100D']
```

```
[75]:      SEQN  BPQ080  BPQ100D
1  109264      NaN      NaN
2  109265      NaN      NaN
3  109266      Yes      NaN
4  109267      No      NaN

```

4.4 Practice

using data tmp for the following task

1. select the middle two rows using loc method
2. select the middle two columns using column name
3. combine 1 and 2 above

```
[76]: tmp = nhane.loc[1:8,
↳ ['SEQN', 'BMXBMI', 'RIDSTATR', 'RIDRETH1', 'BMXLEG', 'BMXWAIST']]
tmp = tmp.set_axis(['A', 'B', 'C', 'D', 'E', 'F', 'G', 'H'], axis='index',)
tmp

```

```
[76]:      SEQN  BMXBMI      RIDSTATR \
A  109264    17.6  Both interviewed and MEC examined
B  109265    15.0  Both interviewed and MEC examined
C  109266    37.8  Both interviewed and MEC examined
D  109267     NaN      Interviewed only
E  109268     NaN      Interviewed only
F  109269     NaN  Both interviewed and MEC examined

```

G	109270	30.9	Both interviewed and MEC examined
H	109271	29.7	Both interviewed and MEC examined

		RIDRETH1	BMXLEG	BMXWAIST
A		Mexican American	36.3	63.8
B		Non-Hispanic White	NaN	41.2
C	Other Race - Including Multi-Racial		40.8	117.9
D		Other Hispanic	NaN	NaN
E		Non-Hispanic White	NaN	NaN
F		Other Hispanic	NaN	NaN
G		Non-Hispanic Black	42.6	91.4
H		Non-Hispanic White	40.1	120.4

```
[77]: tmp.loc[['D','E']]
```

```
[77]:
```

	SEQN	BMXBMI	RIDSTATR	RIDRETH1	BMXLEG	BMXWAIST
D	109267	NaN	Interviewed only	Other Hispanic	NaN	NaN
E	109268	NaN	Interviewed only	Non-Hispanic White	NaN	NaN

```
[78]: tmp.iloc[[3,4]]
```

```
[78]:
```

	SEQN	BMXBMI	RIDSTATR	RIDRETH1	BMXLEG	BMXWAIST
D	109267	NaN	Interviewed only	Other Hispanic	NaN	NaN
E	109268	NaN	Interviewed only	Non-Hispanic White	NaN	NaN

```
[79]: tmp[['RIDSTATR','RIDRETH1']]
```

```
[79]:
```

	RIDSTATR	RIDRETH1
A	Both interviewed and MEC examined	Mexican American
B	Both interviewed and MEC examined	Non-Hispanic White
C	Both interviewed and MEC examined	Other Race - Including Multi-Racial
D	Interviewed only	Other Hispanic
E	Interviewed only	Non-Hispanic White
F	Both interviewed and MEC examined	Other Hispanic
G	Both interviewed and MEC examined	Non-Hispanic Black
H	Both interviewed and MEC examined	Non-Hispanic White

```
[80]: tmp.iloc[[3,4],2:4]
```

```
[80]:
```

	RIDSTATR	RIDRETH1
D	Interviewed only	Other Hispanic
E	Interviewed only	Non-Hispanic White

```
[81]: tmp.loc[['D','E'],['RIDSTATR','RIDRETH1']]
```

```
[81]:
```

	RIDSTATR	RIDRETH1
D	Interviewed only	Other Hispanic
E	Interviewed only	Non-Hispanic White

4.5 Conditonally selecting data

4.5.1 Example 1: Select Rows Where BMXBMI Is Greater Than 25

```
[82]: # Selecting rows where BMI is greater than 25
      nhane[nhane['BMXBMI'] > 25]
```

```
[82]:      Unnamed: 0      SEQN BPQ080 BPQ100D      BMDSTATS      BMXHT  \
3              4  109266      Yes      NaN  Complete data for age group  160.2
7              8  109270      NaN      NaN  Complete data for age group  156.0
8              9  109271      Yes      Yes  Complete data for age group  182.3
11             12  109274      Yes      Yes  Complete data for age group  185.3
19             20  109282      Yes      NaN      Other partial exam  177.1
...
15573          6717  115979      No      No  Complete data for age group  157.5
15576          6720  115982      No      NaN  Complete data for age group  157.2
15577          6721  115983      No      NaN  Complete data for age group  160.8
15579          6723  115985      Yes      Yes  Complete data for age group  173.2
15580          6724  115986      No      NaN  Complete data for age group  164.8

      BMXBMI      BMDMIC      BMXLEG      BMXWAIST  ...      DIQ070      CDQ010      ALQ151      OCQ180  \
3          37.8      NaN      40.8      117.9  ...      No      NaN      No      40.0
7          30.9      Obese      42.6      91.4  ...      NaN      NaN      NaN      NaN
8          29.7      NaN      40.1      120.4  ...      NaN      Yes      Yes      NaN
11         30.2      NaN      44.0      109.6  ...      Yes      No      No      NaN
19         26.6      NaN      39.0      NaN  ...      NaN      Yes      No      NaN
...
15573         42.3      NaN      36.1      115.8  ...      No      NaN      No      30.0
15576         26.1      NaN      32.9      93.5  ...      NaN      NaN      No      36.0
15577         29.5      NaN      38.0      105.6  ...      NaN      Yes      No      NaN
15579         32.0      NaN      44.7      116.7  ...      Yes      No      No      35.0
15580         39.4      NaN      42.0      110.2  ...      NaN      Yes      No      38.0

      LUAXSTAT      LUXCAPM      LBXIN      MCQ053      LBXTC      interview_day
3      Complete      277.0      NaN      No      195.0      2019-04-10
7              NaN      NaN      NaN      Yes      103.0      2018-09-28
8      Complete      303.0      16.96      No      147.0      2017-07-11
11     Complete      279.0      13.52      No      105.0      2018-03-22
19     Complete      315.0      7.49      Yes      233.0      2019-09-07
...
15573     Partial      179.0      NaN      No      196.0      2019-06-29
15576  Ineligible      NaN      10.86      No      134.0      2018-12-19
15577     Complete      328.0      40.32      No      218.0      2017-05-18
15579     Complete      212.0      15.22      No      133.0      2019-11-06
15580     Complete      272.0      NaN      No      225.0      2017-11-23
```

```
[7112 rows x 26 columns]
```

4.5.2 Example 2: Select Rows Where RIDRETH1 (Ethnicity) Is 'Non-Hispanic White'

```
[83]: nhane[nhane['RIDRETH1'] == 'Non-Hispanic White']
```

```
[83]:      Unnamed: 0      SEQN BPQ080 BPQ100D  \
2              3  109265      NaN      NaN
5              6  109268      No      NaN
8              9  109271     Yes     Yes
10             11  109273      No      NaN
12             13  109275      NaN      NaN
...
15566          6710  115972     Yes     Yes
15568          6712  115974      No      NaN
15572          6716  115978      No      NaN
15577          6721  115983      No      NaN
15580          6724  115986      No      NaN

      BMDSTATS  BMXHT  BMXBMI  \
2      Complete data for age group  89.3  15.0
5              NaN      NaN      NaN
8      Complete data for age group  182.3  29.7
10     Complete data for age group  184.2  21.9
12  Partial: Only height and weight obtained  120.4  14.4
...
15566      Complete data for age group  167.0  31.9
15568              NaN      NaN      NaN
15572      Complete data for age group  181.4  33.9
15577      Complete data for age group  160.8  29.5
15580      Complete data for age group  164.8  39.4

      BMDBMIC  BMXLEG  BMXWAIST  ...  DIQ070  CDQ010  ALQ151  OCQ180  \
2      Normal weight      NaN    41.2  ...    NaN    NaN    NaN    NaN
5              NaN      NaN      NaN  ...    NaN    NaN    NaN    NaN
8              NaN    40.1    120.4  ...    NaN    Yes    Yes    NaN
10              NaN    41.0    86.8  ...    NaN    NaN    No    28.0
12      Normal weight      NaN      NaN  ...    No    NaN    NaN    NaN
...
15566          NaN    35.6    115.8  ...    Yes    No    No    NaN
15568          NaN      NaN      NaN  ...    NaN    NaN    NaN    12.0
15572          NaN    44.5    113.0  ...    No    No    NaN    40.0
15577          NaN    38.0    105.6  ...    NaN    Yes    No    NaN
15580          NaN    42.0    110.2  ...    NaN    Yes    No    38.0

      LUAXSTAT  LUXCAPM  LBXIN  MCQ053  LBXTC  interview_day
2              NaN      NaN      NaN    No    NaN    2018-10-08
5              NaN      NaN      NaN    No    NaN    2019-04-07
8      Complete    303.0    16.96    No    147.0    2017-07-11
```

10	Complete	222.0	NaN	No	164.0	2019-01-19
12	NaN	NaN	NaN	No	167.0	2018-11-08
...	...	...	...	...	...	...
15566	Complete	155.0	NaN	No	110.0	2017-03-28
15568	NaN	NaN	NaN	No	NaN	2017-06-28
15572	Complete	290.0	NaN	No	199.0	2018-03-28
15577	Complete	328.0	40.32	No	218.0	2017-05-18
15580	Complete	272.0	NaN	No	225.0	2017-11-23

[5278 rows x 26 columns]

4.5.3 Example 3: Select Rows Where BMXBMI Is Between 18.5 and 24.9 (Normal BMI Range)

```
[84]: nhane[(nhane['BMXBMI'] >= 18.5) & (nhane['BMXBMI'] <= 24.9)]
```

```
[84]: Unnamed: 0    SEQN BPQ080 BPQ100D    BMDSTATS    BMXHT  \
10          11  109273      No      NaN  Complete data for age group  184.2
14          15  109277      NaN      NaN  Complete data for age group  162.0
16          17  109279      No      NaN  Complete data for age group  162.2
24          25  109287      NaN      NaN  Complete data for age group  145.1
32          33  109295      No      NaN  Complete data for age group  157.4
...
15548      15549  124811      No      NaN  Complete data for age group  182.7
15559      15560  124822      Yes     Yes  Complete data for age group  167.5
15567      6711  115973      NaN      NaN  Complete data for age group  112.4
15570      6714  115976      No      NaN  Complete data for age group  150.1
15575      6719  115981      NaN      NaN  Complete data for age group  164.4
```

	BMXBMI	BMDBMIC	BMXLEG	BMXWAIST	...	DIQ070	CDQ010	ALQ151	\
10	21.9	NaN	41.0	86.8	...	NaN	NaN	No	
14	18.6	Normal weight	38.6	67.6	...	NaN	NaN	NaN	
16	21.0	Normal weight	37.6	77.4	...	NaN	NaN	NaN	
24	19.2	Normal weight	35.1	73.5	...	NaN	NaN	NaN	
32	24.9	NaN	34.7	90.5	...	NaN	Yes	NaN	
...	...	...	...	...	...	...	...	...	
15548	20.9	NaN	44.4	90.0	...	NaN	No	No	
15559	21.3	NaN	41.1	86.9	...	No	Yes	Yes	
15567	22.4	Obese	NaN	67.1	...	NaN	NaN	NaN	
15570	24.5	NaN	34.5	79.4	...	NaN	NaN	NaN	
15575	21.2	Normal weight	38.2	79.4	...	NaN	NaN	NaN	

	OCQ180	LUAXSTAT	LUXCAPM	LBXIN	MCQ053	LBXTC	interview_day
10	28.0	Complete	222.0	NaN	No	164.0	2019-01-19
14	NaN	Complete	132.0	6.44	No	129.0	2017-02-28
16	NaN	Complete	235.0	NaN	No	182.0	2018-09-07
24	NaN	NaN	NaN	NaN	No	NaN	2018-05-18

32	43.0	Partial	244.0	NaN	Yes	234.0	2019-11-11
...	...	...	...	...	...	...	...
15548	NaN	Complete	213.0	NaN	No	151.0	2018-08-05
15559	45.0	Partial	100.0	4.45	No	171.0	2018-05-17
15567	NaN	NaN	NaN	NaN	No	NaN	2017-03-23
15570	48.0	Complete	239.0	NaN	No	NaN	2017-08-11
15575	NaN	Complete	185.0	NaN	No	121.0	2019-12-29

[3572 rows x 26 columns]

4.5.4 Example 4: Select Rows Where BMXBMI Is Missing (NaN)

```
[85]: nhane[nhane['BMXBMI'].isna()]
# nhane[nhane['BMXBMI'].isnull()]
```

```
[85]:      Unnamed: 0      SEQN BPQ080 BPQ100D      BMDSTATS BMXHT \
0              1  109263      NaN      NaN      No body measures exam data      NaN
4              5  109267      No      NaN                        NaN      NaN
5              6  109268      No      NaN                        NaN      NaN
6              7  109269      NaN      NaN      Other partial exam      NaN
9             10  109272      NaN      NaN      Complete data for age group      NaN
...
15541      15542  124804      NaN      NaN      Complete data for age group      NaN
15545      15546  124808      No      NaN                        NaN      NaN
15553      15554  124816      NaN      NaN      Complete data for age group      NaN
15561           6705  115967      NaN      NaN      Complete data for age group      NaN
15568           6712  115974      No      NaN                        NaN      NaN
```

	BMXBMI	BMDBMIC	BMXLEG	BMXWAIST	...	DIQ070	CDQ010	ALQ151	OCQ180	\
0	NaN	NaN	NaN	NaN	...	NaN	NaN	NaN	NaN	
4	NaN	NaN	NaN	NaN	...	NaN	NaN	NaN	40.0	
5	NaN	NaN	NaN	NaN	...	NaN	NaN	NaN	NaN	
6	NaN	NaN	NaN	NaN	...	NaN	NaN	NaN	NaN	
9	NaN	NaN	NaN	NaN	...	NaN	NaN	NaN	NaN	
...	...	...	...	...	...	...	...	...	...	
15541	NaN	NaN	NaN	NaN	...	NaN	NaN	NaN	NaN	
15545	NaN	NaN	NaN	NaN	...	NaN	NaN	NaN	36.0	
15553	NaN	NaN	NaN	NaN	...	NaN	NaN	NaN	NaN	
15561	NaN	NaN	NaN	NaN	...	NaN	NaN	NaN	NaN	
15568	NaN	NaN	NaN	NaN	...	NaN	NaN	NaN	12.0	

	LUAXSTAT	LUXCAPM	LBXIN	MCQ053	LBXTC	interview_day
0	NaN	NaN	NaN	No	NaN	2017-10-11
4	NaN	NaN	NaN	No	NaN	2019-03-23
5	NaN	NaN	NaN	No	NaN	2019-04-07
6	NaN	NaN	NaN	No	NaN	2017-02-20
9	NaN	NaN	NaN	NaN	NaN	2019-10-07

```

...      ...      ...      ...      ...
15541      NaN      NaN      NaN      No      NaN      2017-05-11
15545      NaN      NaN      NaN      No      NaN      2019-10-31
15553      NaN      NaN      NaN      No      NaN      2017-02-09
15561      NaN      NaN      NaN      NaN      NaN      2019-10-14
15568      NaN      NaN      NaN      No      NaN      2017-06-28

```

[2425 rows x 26 columns]

4.6 Practice

1. select observation with complete survey status
2. look at observations with taking medicines
3. find observations with bmi over 29
4. select rows with 3, and 5 through 10 (you cannot directly type the number 3,5 until 10)

```
[86]: nhane['RIDSTATR'].unique()
```

```
[86]: array(['Both interviewed and MEC examined', 'Interviewed only'],
      dtype=object)
```

```
[87]: #practise1
nhane[nhane['RIDSTATR']=='Both interviewed and MEC examined']
```

```
[87]:      Unnamed: 0      SEQN BPQ080 BPQ100D      BMDSTATS      BMXHT  \
0           1  109263      NaN      NaN      No body measures exam data      NaN
1           2  109264      NaN      NaN      Complete data for age group      154.7
2           3  109265      NaN      NaN      Complete data for age group      89.3
3           4  109266      Yes      NaN      Complete data for age group      160.2
6           7  109269      NaN      NaN      Other partial exam      NaN
...      ...      ...      ...      ...      ...      ...
15577      6721  115983      No      NaN      Complete data for age group      160.8
15578      6722  115984      NaN      NaN      Complete data for age group      115.9
15579      6723  115985      Yes      Yes      Complete data for age group      173.2
15580      6724  115986      No      NaN      Complete data for age group      164.8
15581      6725  115987      NaN      NaN      Complete data for age group      116.6

```

```

      BMXBMI      BMDBMIC      BMXLEG      BMXWAIST      ...      DIQ070      CDQ010      ALQ151  \
0           NaN           NaN           NaN           NaN      ...      NaN      NaN      NaN
1          17.6      Normal weight          36.3          63.8      ...      NaN      NaN      NaN
2          15.0      Normal weight           NaN          41.2      ...      NaN      NaN      NaN
3          37.8           NaN          40.8          117.9      ...      No      NaN      No
6           NaN           NaN           NaN           NaN      ...      NaN      NaN      NaN
...      ...      ...      ...      ...      ...      ...      ...
15577      29.5           NaN          38.0          105.6      ...      NaN      Yes      No
15578      15.9      Normal weight           NaN          55.5      ...      NaN      NaN      NaN
15579      32.0           NaN          44.7          116.7      ...      Yes      No      No
15580      39.4           NaN          42.0          110.2      ...      NaN      Yes      No

```

15581	14.9	Normal weight	NaN	53.1	...	NaN	NaN	NaN
-------	------	---------------	-----	------	-----	-----	-----	-----

	OCQ180	LUAXSTAT	LUXCAPM	LBXIN	MCQ053	LBXTC	interview_day
0	NaN	NaN	NaN	NaN	No	NaN	2017-10-11
1	NaN	Ineligible	NaN	6.05	No	166.0	2019-01-02
2	NaN	NaN	NaN	NaN	No	NaN	2018-10-08
3	40.0	Complete	277.0	NaN	No	195.0	2019-04-10
6	NaN	NaN	NaN	NaN	No	NaN	2017-02-20
...	...	...	...	...	...	...	...
15577	NaN	Complete	328.0	40.32	No	218.0	2017-05-18
15578	NaN	NaN	NaN	NaN	No	NaN	2017-07-26
15579	35.0	Complete	212.0	15.22	No	133.0	2019-11-06
15580	38.0	Complete	272.0	NaN	No	225.0	2017-11-23
15581	NaN	NaN	NaN	NaN	No	151.0	2018-07-04

[14321 rows x 26 columns]

```
[88]: nhane['BPQ100D'].unique()
```

```
[88]: array([nan, 'Yes', 'No', "Don't know"], dtype=object)
```

```
[89]: nhane[nhane['BPQ100D'] == 'Yes']
```

```
[89]:
```

	Unnamed: 0	SEQN	BPQ080	BPQ100D		BMDSTATS	BMXHT	\
8	9	109271	Yes	Yes	Complete data for age group	182.3		
11	12	109274	Yes	Yes	Complete data for age group	185.3		
20	21	109283	No	Yes		NaN	NaN	
27	28	109290	No	Yes	Complete data for age group	161.2		
35	36	109298	Yes	Yes	Complete data for age group	162.5		
...	...	...	...	...	...	...	...	...
15533	15534	124796	Yes	Yes	Complete data for age group	164.8		
15554	15555	124817	Yes	Yes	Complete data for age group	147.8		
15559	15560	124822	Yes	Yes	Complete data for age group	167.5		
15566	6710	115972	Yes	Yes	Complete data for age group	167.0		
15579	6723	115985	Yes	Yes	Complete data for age group	173.2		

	BMXBMI	BMDBMIC	BMXLEG	BMXWAIST	...	DIQ070	CDQ010	ALQ151	OCQ180	\
8	29.7	NaN	40.1	120.4	...	NaN	Yes	Yes	NaN	
11	30.2	NaN	44.0	109.6	...	Yes	No	No	NaN	
20	NaN	NaN	NaN	NaN	...	NaN	Yes	NaN	NaN	
27	28.1	NaN	38.0	92.0	...	Yes	No	No	60.0	
35	34.2	NaN	38.0	116.2	...	Yes	Yes	Yes	NaN	
...	...	...	...	...	...	...	...	...	...	...
15533	35.6	NaN	36.5	115.7	...	Yes	Yes	NaN	16.0	
15554	37.9	NaN	32.8	110.0	...	Yes	Yes	No	NaN	
15559	21.3	NaN	41.1	86.9	...	No	Yes	Yes	45.0	
15566	31.9	NaN	35.6	115.8	...	Yes	No	No	NaN	

15579	32.0	NaN	44.7	116.7	...	Yes	No	No	35.0
	LUAXSTAT	LUXCAPM	LBXIN	MCQ053	LBXTC	interview_day			
8	Complete	303.0	16.96	No	147.0	2017-07-11			
11	Complete	279.0	13.52	No	105.0	2018-03-22			
20	NaN	NaN	NaN	No	NaN	2017-04-29			
27	Complete	359.0	20.25	No	165.0	2018-07-17			
35	Partial	400.0	NaN	No	158.0	2019-10-09			
...	...	...	...	...	...	...			
15533	Complete	217.0	NaN	No	155.0	2019-11-06			
15554	Complete	357.0	NaN	No	200.0	2018-08-14			
15559	Partial	100.0	4.45	No	171.0	2018-05-17			
15566	Complete	155.0	NaN	No	110.0	2017-03-28			
15579	Complete	212.0	15.22	No	133.0	2019-11-06			

[2009 rows x 26 columns]

5 Introduction to Pandas Data Types

Pandas is a highly versatile library in Python that provides efficient tools for data manipulation and analysis. One of its strengths is the ability to handle a variety of data types within its DataFrame and Series objects, which are central to its functionality. These data types determine how operations are performed on the data and enable pandas to perform optimized computations. The primary data types in pandas include numeric, string, categorical, and datetime types, each of which serves a specific purpose in data analysis.

5.1 1. Numeric Data Types:

Numeric data types are used to store numerical values, either as integers or floating-point numbers (decimals). They are typically used for arithmetic operations, statistical analysis, and mathematical computations.

Common numeric types: int64 (for integers), float64 (for floating-point numbers). Example: Age, salary, temperature. ## 2. String (Object) Data Types: String data in pandas is typically stored as the object data type. This is used to store text data, which can be unstructured or structured, such as categorical values or identifiers.

Common string type: object. Example: Names, product codes, addresses. ## 3. Categorical Data Types: Categorical data types are used for variables that have a limited and fixed number of possible values. These variables represent distinct categories, making this type memory-efficient and faster for certain operations, especially with repeated or low-cardinality data.

Common categorical type: category. Example: Gender (Male/Female), countries, yes/no answers. ## 4. Datetime Data Types: Datetime data types are used to store dates and times. Pandas provides powerful tools to work with datetime data, allowing for operations like date arithmetic, extraction of date components (e.g., year, month, day), and easy handling of time series data.

Common datetime type: datetime64[ns]. Example: Timestamps, event dates, financial time series. Each of these data types is tailored for specific kinds of data and operations, enabling more efficient

data manipulation and analysis. Converting between these types when necessary ensures that pandas performs the correct operations and optimizes memory usage and computational performance.

5.2 5. Missing Data

NAN - Not a number (it is not 0, and it is not space, it's just empty value that we don't know)

5.3 Example of checking and extracting missing values

```
[90]: nhane.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 15582 entries, 0 to 15581
Data columns (total 26 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Unnamed: 0            15582 non-null  int64
1   SEQN                  15582 non-null  int64
2   BPQ080                10207 non-null  object
3   BPQ100D               2751 non-null   object
4   BMDSTATS              14321 non-null  object
5   BMXHT                 13177 non-null  float64
6   BMXBMI                13157 non-null  float64
7   BMDBMIC               4758 non-null   object
8   BMXLEG                10996 non-null  float64
9   BMXWAIST              12593 non-null  float64
10  RIDSTATR              15582 non-null  object
11  RIAGENDR              15582 non-null  object
12  RIDAGEYR              15582 non-null  int64
13  RIDRETH1              15582 non-null  object
14  RIDRETH3              15582 non-null  object
15  RIDEXMON              14321 non-null  object
16  DIQ070                2684 non-null   object
17  CDQ010                6440 non-null   object
18  ALQ151                7509 non-null   object
19  OCQ180                5378 non-null   float64
20  LUAXSTAT              10421 non-null  object
21  LUXCAPM               9708 non-null   float64
22  LBXIN                 4629 non-null   float64
23  MCQ053                15007 non-null  object
24  LBXTC                 10840 non-null  float64
25  interview_day         15582 non-null  datetime64[ns]
dtypes: datetime64[ns](1), float64(8), int64(3), object(14)
memory usage: 3.1+ MB
```

```
[91]: nhane[nhane['BPQ080'].isnull()]
```

```
[91]:      Unnamed: 0      SEQN BPQ080 BPQ100D      BMDSTATS BMXHT \
0          1  109263      NaN      NaN      No body measures exam data      NaN
1          2  109264      NaN      NaN      Complete data for age group      154.7
2          3  109265      NaN      NaN      Complete data for age group      89.3
6          7  109269      NaN      NaN      Other partial exam      NaN
7          8  109270      NaN      NaN      Complete data for age group      156.0
...      ...      ...      ...      ...      ...      ...
15571      6715  115977      NaN      NaN      Complete data for age group      135.9
15574      6718  115980      NaN      NaN      Complete data for age group      89.1
15575      6719  115981      NaN      NaN      Complete data for age group      164.4
15578      6722  115984      NaN      NaN      Complete data for age group      115.9
15581      6725  115987      NaN      NaN      Complete data for age group      116.6
```

```
      BMXBMI      BMDMIC BMXLEG BMXWAIST ... DIQ070 CDQ010 ALQ151 \
0      NaN      NaN      NaN      NaN ...      NaN      NaN      NaN
1      17.6 Normal weight      36.3      63.8 ...      NaN      NaN      NaN
2      15.0 Normal weight      NaN      41.2 ...      NaN      NaN      NaN
6      NaN      NaN      NaN      NaN ...      NaN      NaN      NaN
7      30.9      Obese      42.6      91.4 ...      NaN      NaN      NaN
...      ...      ...      ...      ...      ...      ...
15571      25.7      Obese      32.0      84.2 ...      NaN      NaN      NaN
15574      16.1 Normal weight      NaN      48.0 ...      NaN      NaN      NaN
15575      21.2 Normal weight      38.2      79.4 ...      NaN      NaN      NaN
15578      15.9 Normal weight      NaN      55.5 ...      NaN      NaN      NaN
15581      14.9 Normal weight      NaN      53.1 ...      NaN      NaN      NaN
```

```
      OCQ180      LUAXSTAT LUXCAPM LBXIN MCQ053 LBXTC      interview_day
0      NaN      NaN      NaN      NaN      No      NaN      2017-10-11
1      NaN      Ineligible      NaN      6.05      No      166.0      2019-01-02
2      NaN      NaN      NaN      NaN      No      NaN      2018-10-08
6      NaN      NaN      NaN      NaN      No      NaN      2017-02-20
7      NaN      NaN      NaN      NaN      Yes      103.0      2018-09-28
...      ...      ...      ...      ...      ...
15571      NaN      NaN      NaN      NaN      No      NaN      2017-02-17
15574      NaN      NaN      NaN      NaN      No      NaN      2017-07-29
15575      NaN      Complete      185.0      NaN      No      121.0      2019-12-29
15578      NaN      NaN      NaN      NaN      No      NaN      2017-07-26
15581      NaN      NaN      NaN      NaN      No      151.0      2018-07-04
```

[5375 rows x 26 columns]

```
[92]: nhane[nhane['BPQ080'].isnull()]['BPQ080']
```

```
[92]: 0      NaN
1      NaN
2      NaN
6      NaN
```

```

7         NaN
...
15571     NaN
15574     NaN
15575     NaN
15578     NaN
15581     NaN
Name: BPQ080, Length: 5375, dtype: object

```

6 Basic Datetime Operation

6.1 1. convert text date column using pd.to_datetime method

```
[93]: nhane['interview_day'] = pd.to_datetime(nhane['interview_day'])
```

6.2 2. Extract year, month, day, and day of week

```
[94]: nhane['Day'] = nhane['interview_day'].dt.day
nhane['Month'] = nhane['interview_day'].dt.month
nhane['Year'] = nhane['interview_day'].dt.year
nhane['Dayname'] = nhane['interview_day'].dt.day_name()
nhane['Day_week'] = nhane['interview_day'].dt.day_of_week
```

```
[95]: nhane['Day'] = nhane['interview_day'].dt.day
```

```
[96]: nhane[['SEQN', 'interview_day', 'Day', 'Month', 'Dayname', 'Day_week']]
```

```
[96]:
```

	SEQN	interview_day	Day	Month	Dayname	Day_week
0	109263	2017-10-11	11	10	Wednesday	2
1	109264	2019-01-02	2	1	Wednesday	2
2	109265	2018-10-08	8	10	Monday	0
3	109266	2019-04-10	10	4	Wednesday	2
4	109267	2019-03-23	23	3	Saturday	5
...	...	...	...	...	...	...
15577	115983	2017-05-18	18	5	Thursday	3
15578	115984	2017-07-26	26	7	Wednesday	2
15579	115985	2019-11-06	6	11	Wednesday	2
15580	115986	2017-11-23	23	11	Thursday	3
15581	115987	2018-07-04	4	7	Wednesday	2

```
[15582 rows x 6 columns]
```

6.3 3. date calculation - date difference

```
[97]: nhane['daydiff'] = nhane['interview_day'].diff()  
nhane['daydiff'].head()
```

```
[97]: 0      NaT  
1    448 days  
2    -86 days  
3    184 days  
4    -18 days  
Name: daydiff, dtype: timedelta64[ns]
```

```
[98]: nhane[['interview_day', 'daydiff']].head()  
## Practice
```

```
[98]:   interview_day  daydiff  
0   2017-10-11      NaT  
1   2019-01-02  448 days  
2   2018-10-08  -86 days  
3   2019-04-10  184 days  
4   2019-03-23  -18 days
```

6.4 Practice

1. find the date difference is less than 5 days

```
[99]: nhane[nhane['daydiff'].dt.days<5]['interview_day']
```

```
[99]: 2      2018-10-08  
4      2019-03-23  
6      2017-02-20  
8      2017-07-11  
10     2019-01-19  
  
...  
15571   2017-02-17  
15574   2017-07-29  
15576   2018-12-19  
15577   2017-05-18  
15580   2017-11-23  
Name: interview_day, Length: 7876, dtype: datetime64[ns]
```

6.5 4. Date slicing

1. there is special situation when datetime is in the index we can do date slicing

```
[100]: nhane.set_index(nhane['interview_day'], inplace=True)  
# nhane = nhane.set_index(nhane['interview_day'])  
nhane.sort_index(ascending=True, inplace=True)  
nhane[['SEQN', 'BPQ080', 'BPQ100D', 'BMDSTATS']].head()
```

```
[100]:
```

	SEQN	BPQ080	BPQ100D	BMDSTATS
interview_day				
2017-01-01	112413	NaN	NaN	Complete data for age group
2017-01-01	114846	Yes	Yes	Other partial exam
2017-01-01	110026	NaN	NaN	Complete data for age group
2017-01-01	120431	No	NaN	Complete data for age group
2017-01-01	116500	No	NaN	Complete data for age group

```
[101]: nhane.loc['2017-01':'2017-05']
```

```
[101]:
```

	Unnamed: 0	SEQN	BPQ080	BPQ100D	\
interview_day					
2017-01-01	3151	112413	NaN	NaN	
2017-01-01	5584	114846	Yes	Yes	
2017-01-01	764	110026	NaN	NaN	
2017-01-01	11169	120431	No	NaN	
2017-01-01	7238	116500	No	NaN	
...	...	...	...	...	
2017-05-31	4789	114051	NaN	NaN	
2017-05-31	15045	124307	NaN	NaN	
2017-05-31	14578	123840	No	NaN	
2017-05-31	4310	113572	NaN	NaN	
2017-05-31	11944	121206	No	NaN	

	BMDSTATS	BMXHT	BMXBMI	\
interview_day				
2017-01-01	Complete data for age group	163.9	15.6	
2017-01-01	Other partial exam	174.3	34.6	
2017-01-01	Complete data for age group	125.9	15.8	
2017-01-01	Complete data for age group	178.2	34.8	
2017-01-01	Complete data for age group	166.7	38.9	
...	...	...	...	
2017-05-31	Complete data for age group	127.3	16.3	
2017-05-31	Complete data for age group	164.4	17.8	
2017-05-31	Complete data for age group	164.6	25.0	
2017-05-31	Complete data for age group	118.7	15.5	
2017-05-31	Partial: Only height and weight obtained	172.6	19.4	

	BMDBMIC	BMXLEG	BMXWAIST	... LBXIN	MCQ053	LBXTC	\
interview_day				...			
2017-01-01	Normal weight	39.2	65.4	...	NaN	No	NaN
2017-01-01	NaN	39.5	NaN	...	NaN	No	181.0
2017-01-01	Normal weight	NaN	56.3	...	NaN	No	168.0
2017-01-01	Obese	45.2	117.3	...	NaN	No	NaN
2017-01-01	NaN	37.5	125.0	...	9.04	No	144.0
...	...	...	...	...	...		
2017-05-31	Normal weight	31.0	54.2	...	NaN	No	149.0

2017-05-31	Normal weight	40.7	64.2	...	NaN	No	175.0
2017-05-31	NaN	34.2	85.9	...	NaN	No	166.0
2017-05-31	Normal weight	NaN	58.6	...	NaN	No	155.0
2017-05-31	Normal weight	NaN	NaN	...	NaN	No	NaN

	interview_day	Day	Month	Year	Dayname	Day_week	daydiff
interview_day							
2017-01-01	2017-01-01	1	1	2017	Sunday	6	-266 days
2017-01-01	2017-01-01	1	1	2017	Sunday	6	-161 days
2017-01-01	2017-01-01	1	1	2017	Sunday	6	-484 days
2017-01-01	2017-01-01	1	1	2017	Sunday	6	-143 days
2017-01-01	2017-01-01	1	1	2017	Sunday	6	-1054 days
...	...	..	...	...	...	...	...
2017-05-31	2017-05-31	31	5	2017	Wednesday	2	-230 days
2017-05-31	2017-05-31	31	5	2017	Wednesday	2	-614 days
2017-05-31	2017-05-31	31	5	2017	Wednesday	2	-57 days
2017-05-31	2017-05-31	31	5	2017	Wednesday	2	-79 days
2017-05-31	2017-05-31	31	5	2017	Wednesday	2	-553 days

[2160 rows x 32 columns]

6.6 Practice

1. Extract the 3rd observation of each day
2. Extract data in August

```
[102]: nhane.groupby(nhane.index).nth(3)
```

```
[102]:      Unnamed: 0      SEQN BPQ080 BPQ100D      BMDSTATS \
interview_day
2017-01-01      11169  120431      No      NaN  Complete data for age group
2017-01-02      13347  122609      No      NaN  Complete data for age group
2017-01-03      9563  118825      NaN      NaN  Complete data for age group
2017-01-04      1529  110791      No      NaN  Complete data for age group
2017-01-05      1222  110484      NaN      NaN  Complete data for age group
...
2019-12-27      12166  121428      Yes      NaN  Complete data for age group
2019-12-28      3809  113071      NaN      NaN  Complete data for age group
2019-12-29      5137  114399      NaN      NaN  Complete data for age group
2019-12-30      3088  112350      No      NaN  Complete data for age group
2019-12-31      10689  119951      No      NaN  Complete data for age group

      BMXHT  BMXBMI      BMDMIC  BMXLEG  BMXWAIST  ...  LBXIN  MCQ053  \
interview_day
2017-01-01  178.2   34.8      Obese   45.2    117.3  ...   NaN    No
2017-01-02  159.4   20.7      NaN    37.0    77.0  ...   2.77  No
2017-01-03  148.6   22.6  Overweight  39.2    74.4  ...   NaN    No
```

2017-01-04	152.2	24.5		NaN	35.0	86.5	...	NaN	No
2017-01-05	135.8	20.2	Overweight		32.5	69.2	...	NaN	No
...	...	...	...	...	...	...	...	...	...
2019-12-27	177.8	26.7		NaN	44.0	93.8	...	NaN	No
2019-12-28	NaN	NaN		NaN	NaN	NaN	...	NaN	NaN
2019-12-29	142.8	21.1	Overweight		33.8	72.7	...	NaN	No
2019-12-30	149.3	40.0		NaN	33.3	110.3	...	NaN	Yes
2019-12-31	168.4	27.6		NaN	41.6	102.1	...	11.21	No

	LBXTC	interview_day	Day	Month	Year	Dayname	Day_week	\
interview_day								
2017-01-01	NaN	2017-01-01	1	1	2017	Sunday	6	
2017-01-02	178.0	2017-01-02	2	1	2017	Monday	0	
2017-01-03	132.0	2017-01-03	3	1	2017	Tuesday	1	
2017-01-04	248.0	2017-01-04	4	1	2017	Wednesday	2	
2017-01-05	NaN	2017-01-05	5	1	2017	Thursday	3	
...	...	...	...	...	...	...	...	...
2019-12-27	185.0	2019-12-27	27	12	2019	Friday	4	
2019-12-28	NaN	2019-12-28	28	12	2019	Saturday	5	
2019-12-29	113.0	2019-12-29	29	12	2019	Sunday	6	
2019-12-30	179.0	2019-12-30	30	12	2019	Monday	0	
2019-12-31	129.0	2019-12-31	31	12	2019	Tuesday	1	

	daydiff
interview_day	
2017-01-01	-143 days
2017-01-02	-251 days
2017-01-03	-528 days
2017-01-04	-995 days
2017-01-05	-829 days
...	...
2019-12-27	304 days
2019-12-28	445 days
2019-12-29	5 days
2019-12-30	10 days
2019-12-31	909 days

[1093 rows x 32 columns]

```
[103]: nhane.loc['2017-08']
```

	Unnamed: 0	SEQN	BPQ080	BPQ100D	BMDSTATS	\
interview_day						
2017-08-01	7504	116766	No	Yes	Complete data for age group	
2017-08-01	5728	114990	No	NaN	Complete data for age group	
2017-08-01	9067	118329	NaN	NaN	Complete data for age group	
2017-08-01	11572	120834	No	NaN	No body measures exam data	

2017-08-01	1416	110678	No	NaN	Complete data for age group
...	...	...	...	...	...
2017-08-31	5214	114476	NaN	NaN	Complete data for age group
2017-08-31	8401	117663	NaN	NaN	Complete data for age group
2017-08-31	7799	117061	No	NaN	Complete data for age group
2017-08-31	3155	112417	NaN	NaN	Complete data for age group
2017-08-31	10881	120143	No	NaN	NaN

	BMXHT	BMXBMI	BMDBMIC	BMXLEG	BMXWAIST	...	LBXIN	\
interview_day						...		
2017-08-01	164.2	31.4	NaN	39.7	105.0	...	12.35	
2017-08-01	173.9	34.9	NaN	39.0	117.6	...	NaN	
2017-08-01	172.8	22.7	Overweight	42.9	86.3	...	NaN	
2017-08-01	NaN	NaN	NaN	NaN	NaN	...	6.11	
2017-08-01	186.8	19.3	NaN	43.0	79.6	...	NaN	
...	...	...	...	...	...	...		
2017-08-31	133.4	17.0	Normal weight	31.3	55.3	...	NaN	
2017-08-31	112.8	19.3	Obese	NaN	60.8	...	NaN	
2017-08-31	172.5	28.4	NaN	45.7	92.7	...	NaN	
2017-08-31	146.0	16.1	Normal weight	35.3	61.8	...	12.03	
2017-08-31	NaN	NaN	NaN	NaN	NaN	...	NaN	

	MCQ053	LBXTC	interview_day	Day	Month	Year	Dayname	Day_week	\
interview_day									
2017-08-01	No	149.0	2017-08-01	1	8	2017	Tuesday	1	
2017-08-01	No	NaN	2017-08-01	1	8	2017	Tuesday	1	
2017-08-01	No	147.0	2017-08-01	1	8	2017	Tuesday	1	
2017-08-01	No	207.0	2017-08-01	1	8	2017	Tuesday	1	
2017-08-01	No	156.0	2017-08-01	1	8	2017	Tuesday	1	
...	...	...	...	...	...	...	...		
2017-08-31	No	NaN	2017-08-31	31	8	2017	Thursday	3	
2017-08-31	No	NaN	2017-08-31	31	8	2017	Thursday	3	
2017-08-31	No	174.0	2017-08-31	31	8	2017	Thursday	3	
2017-08-31	No	79.0	2017-08-31	31	8	2017	Thursday	3	
2017-08-31	No	NaN	2017-08-31	31	8	2017	Thursday	3	

	daydiff
interview_day	
2017-08-01	-490 days
2017-08-01	-376 days
2017-08-01	-561 days
2017-08-01	-814 days
2017-08-01	-172 days
...	...
2017-08-31	-778 days
2017-08-31	-500 days
2017-08-31	-801 days

```
2017-08-31    138 days
2017-08-31    -39 days
```

```
[456 rows x 32 columns]
```

```
[104]: nhane.head()
```

```
[104]:      Unnamed: 0      SEQN BPQ080 BPQ100D      BMDSTATS \
interview_day
2017-01-01      3151  112413      NaN      NaN  Complete data for age group
2017-01-01      5584  114846      Yes      Yes      Other partial exam
2017-01-01       764  110026      NaN      NaN  Complete data for age group
2017-01-01     11169  120431      No      NaN  Complete data for age group
2017-01-01      7238  116500      No      NaN  Complete data for age group

      BMXHT  BMXBMI      BMDBMIC  BMXLEG  BMXWAIST  ...  LBXIN  \
interview_day
2017-01-01  163.9   15.6  Normal weight   39.2     65.4  ...   NaN
2017-01-01  174.3   34.6           NaN   39.5     NaN  ...   NaN
2017-01-01  125.9   15.8  Normal weight   NaN     56.3  ...   NaN
2017-01-01  178.2   34.8         Obese   45.2    117.3  ...   NaN
2017-01-01  166.7   38.9           NaN   37.5    125.0  ...  9.04

      MCQ053  LBXTC  interview_day Day Month  Year Dayname Day_week  \
interview_day
2017-01-01      No   NaN   2017-01-01   1    1  2017  Sunday        6
2017-01-01      No  181.0   2017-01-01   1    1  2017  Sunday        6
2017-01-01      No  168.0   2017-01-01   1    1  2017  Sunday        6
2017-01-01      No   NaN   2017-01-01   1    1  2017  Sunday        6
2017-01-01      No  144.0   2017-01-01   1    1  2017  Sunday        6

      daydiff
interview_day
2017-01-01   -266 days
2017-01-01   -161 days
2017-01-01   -484 days
2017-01-01   -143 days
2017-01-01  -1054 days
```

```
[5 rows x 32 columns]
```

7 Data Cleaning

1. introduce simple method: head, tail, unique, info, describe, sample, shape, rename, sort_value (sort_index)
2. identify duplicate observations/meaningless columns (drop, drop_duplicate)

3. identify outliers
4. impute missing value

— Above is a must —

5. groupby method
6. cross tab method
7. data transpose method
8. apply method
9. test-train split

7.1 Some simple Methods

7.1.1 head, tail, and sample

```
[105]: nhane.head(n=20) # first 10 observations default is 5
```

```
[105]:
```

	Unnamed: 0	SEQN	BPQ080	BPQ100D	BMDSTATS	\
interview_day						
2017-01-01	3151	112413	NaN	NaN	Complete data for age group	
2017-01-01	5584	114846	Yes	Yes	Other partial exam	
2017-01-01	764	110026	NaN	NaN	Complete data for age group	
2017-01-01	11169	120431	No	NaN	Complete data for age group	
2017-01-01	7238	116500	No	NaN	Complete data for age group	
2017-01-01	10038	119300	No	NaN	Complete data for age group	
2017-01-01	1363	110625	No	NaN	Complete data for age group	
2017-01-01	12376	121638	No	NaN	Complete data for age group	
2017-01-01	13780	123042	No	NaN	Complete data for age group	
2017-01-01	15307	124569	No	NaN	Complete data for age group	
2017-01-01	3495	112757	NaN	NaN	Complete data for age group	
2017-01-01	13718	122980	NaN	NaN	Complete data for age group	
2017-01-01	11495	120757	No	NaN	Complete data for age group	
2017-01-01	14315	123577	Yes	No	Complete data for age group	
2017-01-01	15441	124703	No	Yes	NaN	
2017-01-01	13902	123164	No	NaN	Complete data for age group	
2017-01-02	13990	123252	Yes	NaN	Complete data for age group	
2017-01-02	6744	116006	NaN	NaN	Complete data for age group	
2017-01-02	12001	121263	NaN	NaN	Complete data for age group	
2017-01-02	13347	122609	No	NaN	Complete data for age group	

	BMXHT	BMXBMI	BMDBMIC	BMXLEG	BMXWAIST	... LBXIN	\
interview_day							
2017-01-01	163.9	15.6	Normal weight	39.2	65.4	...	NaN
2017-01-01	174.3	34.6	NaN	39.5	NaN	...	NaN
2017-01-01	125.9	15.8	Normal weight	NaN	56.3	...	NaN
2017-01-01	178.2	34.8	Obese	45.2	117.3	...	NaN

2017-01-01	166.7	38.9	NaN	37.5	125.0	...	9.04
2017-01-01	160.6	26.8	NaN	37.2	92.5	...	NaN
2017-01-01	169.0	36.6	NaN	46.4	111.1	...	NaN
2017-01-01	158.2	28.8	NaN	39.8	93.9	...	NaN
2017-01-01	159.4	33.5	NaN	39.0	102.0	...	NaN
2017-01-01	157.6	23.5	NaN	39.4	78.1	...	6.50
2017-01-01	116.6	16.0	Normal weight	NaN	52.0	...	NaN
2017-01-01	185.5	19.2	Normal weight	44.2	68.6	...	NaN
2017-01-01	160.4	35.2	Obese	38.4	107.3	...	NaN
2017-01-01	175.3	41.7	NaN	40.1	130.7	...	NaN
2017-01-01	NaN	NaN	NaN	NaN	NaN	...	NaN
2017-01-01	175.4	23.0	NaN	38.7	83.4	...	5.88
2017-01-02	162.7	35.1	NaN	36.1	115.8	...	NaN
2017-01-02	156.3	19.9	Normal weight	39.5	76.1	...	NaN
2017-01-02	142.3	25.6	Obese	34.6	84.5	...	NaN
2017-01-02	159.4	20.7	NaN	37.0	77.0	...	2.77

	MCQ053	LBXTC	interview_day	Day	Month	Year	Dayname	Day_week \
interview_day								
2017-01-01	No	NaN	2017-01-01	1	1	2017	Sunday	6
2017-01-01	No	181.0	2017-01-01	1	1	2017	Sunday	6
2017-01-01	No	168.0	2017-01-01	1	1	2017	Sunday	6
2017-01-01	No	NaN	2017-01-01	1	1	2017	Sunday	6
2017-01-01	No	144.0	2017-01-01	1	1	2017	Sunday	6
2017-01-01	No	173.0	2017-01-01	1	1	2017	Sunday	6
2017-01-01	No	237.0	2017-01-01	1	1	2017	Sunday	6
2017-01-01	No	160.0	2017-01-01	1	1	2017	Sunday	6
2017-01-01	No	173.0	2017-01-01	1	1	2017	Sunday	6
2017-01-01	No	106.0	2017-01-01	1	1	2017	Sunday	6
2017-01-01	No	190.0	2017-01-01	1	1	2017	Sunday	6
2017-01-01	No	159.0	2017-01-01	1	1	2017	Sunday	6
2017-01-01	No	NaN	2017-01-01	1	1	2017	Sunday	6
2017-01-01	No	197.0	2017-01-01	1	1	2017	Sunday	6
2017-01-01	No	NaN	2017-01-01	1	1	2017	Sunday	6
2017-01-01	No	173.0	2017-01-01	1	1	2017	Sunday	6
2017-01-02	No	208.0	2017-01-02	2	1	2017	Monday	0
2017-01-02	No	NaN	2017-01-02	2	1	2017	Monday	0
2017-01-02	No	143.0	2017-01-02	2	1	2017	Monday	0
2017-01-02	No	178.0	2017-01-02	2	1	2017	Monday	0

daydiff	
interview_day	
2017-01-01	-266 days
2017-01-01	-161 days
2017-01-01	-484 days
2017-01-01	-143 days
2017-01-01	-1054 days

```

2017-01-01    -1019 days
2017-01-01     -137 days
2017-01-01      -29 days
2017-01-01    -321 days
2017-01-01    -526 days
2017-01-01    -696 days
2017-01-01    -781 days
2017-01-01    -679 days
2017-01-01    -800 days
2017-01-01    -659 days
2017-01-01    -619 days
2017-01-02      -2 days
2017-01-02   -1048 days
2017-01-02    -236 days
2017-01-02    -251 days

```

[20 rows x 32 columns]

```
[106]: nhane.tail() # last 10 observations default is 5
```

```

[106]:      Unnamed: 0      SEQN BPQ080 BPQ100D      BMDSTATS \
interview_day
2019-12-31      13504  122766    Yes    Yes  Complete data for age group
2019-12-31      12531  121793     No   NaN  Complete data for age group
2019-12-31      14487  123749    Yes    No  Complete data for age group
2019-12-31       5615  114877     No   NaN  Complete data for age group
2019-12-31       8281  117543     No   NaN  Complete data for age group

      BMXHT  BMXBMI  BMDBMIC  BMXLEG  BMXWAIST  ...  LBXIN  MCQ053  \
interview_day
2019-12-31  172.3    25.8    NaN    40.0    104.2  ...   NaN    No
2019-12-31  154.6    33.5    NaN    38.5     91.5  ...  11.97    No
2019-12-31  180.5    36.3    NaN    41.4    117.4  ...   NaN    No
2019-12-31  171.3    32.6    NaN    38.5    100.5  ...  25.75    No
2019-12-31  176.0    19.9    NaN    41.1     77.4  ...   NaN    No

      LBXTC  interview_day  Day  Month  Year  Dayname  Day_week  daydiff
interview_day
2019-12-31  198.0    2019-12-31  31    12  2019  Tuesday        1    84 days
2019-12-31  152.0    2019-12-31  31    12  2019  Tuesday        1   179 days
2019-12-31  203.0    2019-12-31  31    12  2019  Tuesday        1   979 days
2019-12-31  183.0    2019-12-31  31    12  2019  Tuesday        1   226 days
2019-12-31  119.0    2019-12-31  31    12  2019  Tuesday        1   418 days

```

[5 rows x 32 columns]

```
[107]: nhane.sample(n=5) # default is 1
```

```
[107]: Unnamed: 0      SEQN BPQ080 BPQ100D      BMDSTATS \
interview_day
2017-07-25      2309  111571      No      NaN  Complete data for age group
2018-02-10      7288  116550      NaN      NaN  Complete data for age group
2018-02-01      6912  116174      Yes      Yes  Complete data for age group
2017-03-26      4678  113940      NaN      NaN  Complete data for age group
2017-08-09     14584  123846      No      NaN  Complete data for age group

      BMXHT  BMXBMI      BMDBMIC  BMXLEG  BMXWAIST  ...  LBXIN  \
interview_day
2017-07-25    155.5    33.0      NaN    31.5    116.3  ...    NaN
2018-02-10    143.3    34.7      Obese   35.0    96.5  ...    NaN
2018-02-01    145.6    28.5      NaN    33.5    100.6  ...  10.61
2017-03-26     96.7    15.9  Normal weight   NaN    50.4  ...    NaN
2017-08-09    150.9    24.4      NaN    36.5    78.3  ...    6.74

      MCQ053  LBXTC  interview_day Day Month  Year  Dayname Day_week  \
interview_day
2017-07-25      No  221.0    2017-07-25  25    7  2017    Tuesday        1
2018-02-10      No   NaN    2018-02-10  10    2  2018    Saturday        5
2018-02-01      No  165.0    2018-02-01   1    2  2018    Thursday        3
2017-03-26      No   NaN    2017-03-26  26    3  2017     Sunday        6
2017-08-09     Yes  137.0    2017-08-09   9    8  2017   Wednesday        2

      daydiff
interview_day
2017-07-25  -694 days
2018-02-10    0 days
2018-02-01  -555 days
2017-03-26  -335 days
2017-08-09  -139 days

[5 rows x 32 columns]
```

7.1.2 shape, info, and describe, unique

```
[108]: nhane.shape
```

```
[108]: (15582, 32)
```

```
[109]: nhane.info()
```

```
<class 'pandas.core.frame.DataFrame'>
DatetimeIndex: 15582 entries, 2017-01-01 to 2019-12-31
Data columns (total 32 columns):
#   Column      Non-Null Count  Dtype
---  -
0   Unnamed: 0    15582 non-null  int64
```

```

1  SEQN          15582 non-null int64
2  BPQ080        10207 non-null object
3  BPQ100D       2751 non-null object
4  BMDSTATS      14321 non-null object
5  BMXHT         13177 non-null float64
6  BMXBMI        13157 non-null float64
7  BMDBMIC       4758 non-null object
8  BMXLEG        10996 non-null float64
9  BMXWAIST      12593 non-null float64
10 RIDSTATR      15582 non-null object
11 RIAGENDR      15582 non-null object
12 RIDAGEYR      15582 non-null int64
13 RIDRETH1      15582 non-null object
14 RIDRETH3      15582 non-null object
15 RIDEXMON      14321 non-null object
16 DIQ070        2684 non-null object
17 CDQ010        6440 non-null object
18 ALQ151        7509 non-null object
19 OCQ180        5378 non-null float64
20 LUAXSTAT      10421 non-null object
21 LUXCAPM       9708 non-null float64
22 LBXIN         4629 non-null float64
23 MCQ053        15007 non-null object
24 LBXTC         10840 non-null float64
25 interview_day 15582 non-null datetime64[ns]
26 Day           15582 non-null int32
27 Month         15582 non-null int32
28 Year          15582 non-null int32
29 Dayname       15582 non-null object
30 Day_week      15582 non-null int32
31 daydiff       15581 non-null timedelta64[ns]
dtypes: datetime64[ns](1), float64(8), int32(4), int64(3), object(15),
timedelta64[ns](1)
memory usage: 4.2+ MB

```

```
[110]: nhane.describe().T
```

```

[110]:

```

	count	mean	min \
Unnamed: 0	15582.0	7778.99493	1.0
SEQN	15582.0	117040.99493	109263.0
BMXHT	13177.0	156.466821	78.3
BMXBMI	13157.0	26.655362	11.9
BMXLEG	10996.0	38.74468	24.8
BMXWAIST	12593.0	89.659104	40.0
RIDAGEYR	15582.0	33.735721	0.0
OCQ180	5378.0	193.918929	5.0
LUXCAPM	9708.0	257.571178	100.0

LBXIN	4629.0		14.97919		0.71
LBXTC	10840.0		177.456827		71.0
interview_day	15582	2018-06-30 11:38:33.592606720		2017-01-01 00:00:00	
Day	15582.0		15.72956		1.0
Month	15582.0		6.52766		1.0
Year	15582.0		2017.9957		2017.0
Day_week	15582.0		3.025542		0.0
daydiff	15581	0 days 00:24:35.027276811		-1080 days +00:00:00	

		25%		50%		75%	\
Unnamed: 0		3896.25		7769.5		11664.75	
SEQN		113158.25		117031.5		120926.75	
BMXHT		151.1		162.0		171.3	
BMXBMI		20.4		25.8		31.4	
BMXLEG		36.0		38.9		41.5	
BMXWAIST		73.3		91.0		105.4	
RIDAGEYR		10.0		30.0		56.0	
OCQ180		31.0		40.0		46.0	
LUXCAPM		211.0		253.0		302.0	
LBXIN		6.31		10.18		16.61	
LBXTC		149.0		173.0		201.0	
interview_day	2017-09-26 00:00:00		2018-07-02 00:00:00		2019-04-01 18:00:00		
Day		8.0		16.0		23.0	
Month		4.0		7.0		9.0	
Year		2017.0		2018.0		2019.0	
Day_week		1.0		3.0		5.0	
daydiff	-316 days +00:00:00		-2 days +00:00:00		323 days 00:00:00		

		max		std
Unnamed: 0		15560.0		4488.935536
SEQN		124822.0		4488.935536
BMXHT		199.6		22.639481
BMXBMI		92.3		8.41986
BMXLEG		55.0		4.132493
BMXWAIST		187.5		23.156093
RIDAGEYR		80.0		25.322085
OCQ180		99999.0		3807.572773
LUXCAPM		400.0		63.545593
LBXIN		512.5		22.838416
LBXTC		446.0		40.363195
interview_day	2019-12-31 00:00:00			NaN
Day		31.0		8.799897
Month		12.0		3.444757
Year		2019.0		0.814977
Day_week		6.0		2.011102
daydiff	1081 days 00:00:00		446 days 00:33:01.069954008	

```
[111]: nhane['RIDSTATR'].unique()
```

```
[111]: array(['Both interviewed and MEC examined', 'Interviewed only'],  
        dtype=object)
```

7.1.3 rename, sort_value/sort_index

```
[112]: nhane.rename({'interview_day': 'interview_date'}, inplace=True, axis=1)  
nhane.info()
```

```
<class 'pandas.core.frame.DataFrame'>  
DatetimeIndex: 15582 entries, 2017-01-01 to 2019-12-31  
Data columns (total 32 columns):  
#   Column                Non-Null Count  Dtype  
---  ---  
0   Unnamed: 0            15582 non-null  int64  
1   SEQN                  15582 non-null  int64  
2   BPQ080                10207 non-null  object  
3   BPQ100D               2751 non-null  object  
4   BMDSTATS              14321 non-null  object  
5   BMXHT                 13177 non-null  float64  
6   BMXBMI                13157 non-null  float64  
7   BMDBMIC               4758 non-null  object  
8   BMXLEG                10996 non-null  float64  
9   BMXWAIST              12593 non-null  float64  
10  RIDSTATR              15582 non-null  object  
11  RIAGENDR              15582 non-null  object  
12  RIDAGEYR              15582 non-null  int64  
13  RIDRETH1              15582 non-null  object  
14  RIDRETH3              15582 non-null  object  
15  RIDEXMON              14321 non-null  object  
16  DIQ070                2684 non-null  object  
17  CDQ010                6440 non-null  object  
18  ALQ151                7509 non-null  object  
19  OCQ180                5378 non-null  float64  
20  LUAXSTAT              10421 non-null  object  
21  LUXCAPM               9708 non-null  float64  
22  LBXIN                 4629 non-null  float64  
23  MCQ053                15007 non-null  object  
24  LBXTC                 10840 non-null  float64  
25  interview_date        15582 non-null  datetime64[ns]  
26  Day                   15582 non-null  int32  
27  Month                 15582 non-null  int32  
28  Year                  15582 non-null  int32  
29  Dayname               15582 non-null  object  
30  Day_week              15582 non-null  int32  
31  daydiff               15581 non-null  timedelta64[ns]  
dtypes: datetime64[ns](1), float64(8), int32(4), int64(3), object(15),
```

```
timedelta64[ns](1)
memory usage: 4.2+ MB
```

```
[113]: nahane = nhane.sort_values('SEQN',ascending=False) # this is a view

# following can overwrite the data (choose one of them is fine)

# nhane.sort_values('SEQN',ascending=True,inplace=True)
# nhane = nhane.sort_values('SEQN',ascending=True)

# same for sorting index
```

```
[114]: nhane.head()
```

```
[114]:
```

	Unnamed: 0	SEQN	BPQ080	BPQ100D	BMDSTATS	\
interview_day						
2017-01-01	3151	112413	NaN	NaN	Complete data for age group	
2017-01-01	5584	114846	Yes	Yes	Other partial exam	
2017-01-01	764	110026	NaN	NaN	Complete data for age group	
2017-01-01	11169	120431	No	NaN	Complete data for age group	
2017-01-01	7238	116500	No	NaN	Complete data for age group	

	BMXHT	BMXBMI	BMDBMIC	BMXLEG	BMXWAIST	...	LBXIN	\
interview_day						...		
2017-01-01	163.9	15.6	Normal weight	39.2	65.4	...	NaN	
2017-01-01	174.3	34.6	NaN	39.5	NaN	...	NaN	
2017-01-01	125.9	15.8	Normal weight	NaN	56.3	...	NaN	
2017-01-01	178.2	34.8	Obese	45.2	117.3	...	NaN	
2017-01-01	166.7	38.9	NaN	37.5	125.0	...	9.04	

	MCQ053	LBXTC	interview_date	Day	Month	Year	Dayname	Day_week	\
interview_day									
2017-01-01	No	NaN	2017-01-01	1	1	2017	Sunday	6	
2017-01-01	No	181.0	2017-01-01	1	1	2017	Sunday	6	
2017-01-01	No	168.0	2017-01-01	1	1	2017	Sunday	6	
2017-01-01	No	NaN	2017-01-01	1	1	2017	Sunday	6	
2017-01-01	No	144.0	2017-01-01	1	1	2017	Sunday	6	


```

daydiff
interview_day
2017-01-01    -266 days
2017-01-01    -161 days
2017-01-01    -484 days
2017-01-01    -143 days
2017-01-01   -1054 days

[5 rows x 32 columns]
```

7.2 Duplicate value, meaningless column, and NaNs

```
[115]: nhane.drop(columns = ['Unnamed: 0'],axis=1, inplace=True)
```

```
[116]: # duplicate values
cols = list(set(nhane.columns).difference(set(['interview_date', 'Day',
                                                'Month', 'Year', 'Dayname',
                                                'Day_week', 'daydiff'])))
nhane[nhane.duplicated(subset=cols)].head()
```

```
[116]:
```

	SEQN	BPQ080	BPQ100D		BMDSTATS	BMXHT	\
interview_day							
2018-03-28	115978	No	NaN	Complete data for age group	181.4		
2018-04-01	115984	NaN	NaN	Complete data for age group	115.9		
2018-04-20	115976	No	NaN	Complete data for age group	150.1		
2018-05-23	115968	No	NaN	Complete data for age group	164.3		
2018-07-04	115987	NaN	NaN	Complete data for age group	116.6		

	BMXBMI		BMDBMIC	BMXLEG	BMXWAIST	\
interview_day						
2018-03-28	33.9		NaN	44.5	113.0	
2018-04-01	15.9	Normal weight		NaN	55.5	
2018-04-20	24.5		NaN	34.5	79.4	
2018-05-23	25.2		NaN	36.1	98.0	
2018-07-04	14.9	Normal weight		NaN	53.1	

		RIDSTATR	...	LBXIN	MCQ053	LBXTC	\
interview_day							
2018-03-28	Both interviewed and MEC examined	...	NaN	No	199.0		
2018-04-01	Both interviewed and MEC examined	...	NaN	No	NaN		
2018-04-20	Both interviewed and MEC examined	...	NaN	No	NaN		
2018-05-23	Both interviewed and MEC examined	...	9.73	No	206.0		
2018-07-04	Both interviewed and MEC examined	...	NaN	No	151.0		

	interview_date	Day	Month	Year	Dayname	Day_week	daydiff
interview_day							
2018-03-28	2018-03-28	28	3	2018	Wednesday	2	404 days
2018-04-01	2018-04-01	1	4	2018	Sunday	6	-136 days
2018-04-20	2018-04-20	20	4	2018	Friday	4	-198 days
2018-05-23	2018-05-23	23	5	2018	Wednesday	2	-413 days
2018-07-04	2018-07-04	4	7	2018	Wednesday	2	223 days

[5 rows x 31 columns]

```
[117]: nhane.drop_duplicates(subset=cols,inplace=True)
```

7.3 Outliers detection (use some plot method to visualize)

1. by common sense/expert
2. by algorithm (<https://github.com/yzhao062/pyod>)

7.3.1 Use plotting method and expert

```
[118]: nhane[['BMXBMI', 'BMXLEG', 'BMXWAIST', 'BMXHT']].describe().T
```

```
# max = 92 this observation is an outlier
```

```
[118]:
```

	count	mean	std	min	25%	50%	75%	max
BMXBMI	13137.0	26.656847	8.419842	11.9	20.4	25.8	31.4	92.3
BMXLEG	10984.0	38.745666	4.132524	24.8	36.0	38.9	41.5	55.0
BMXWAIST	12574.0	89.669135	23.150206	40.0	73.3	91.0	105.4	187.5
BMXHT	13157.0	156.490005	22.621786	78.3	151.1	162.1	171.3	199.6

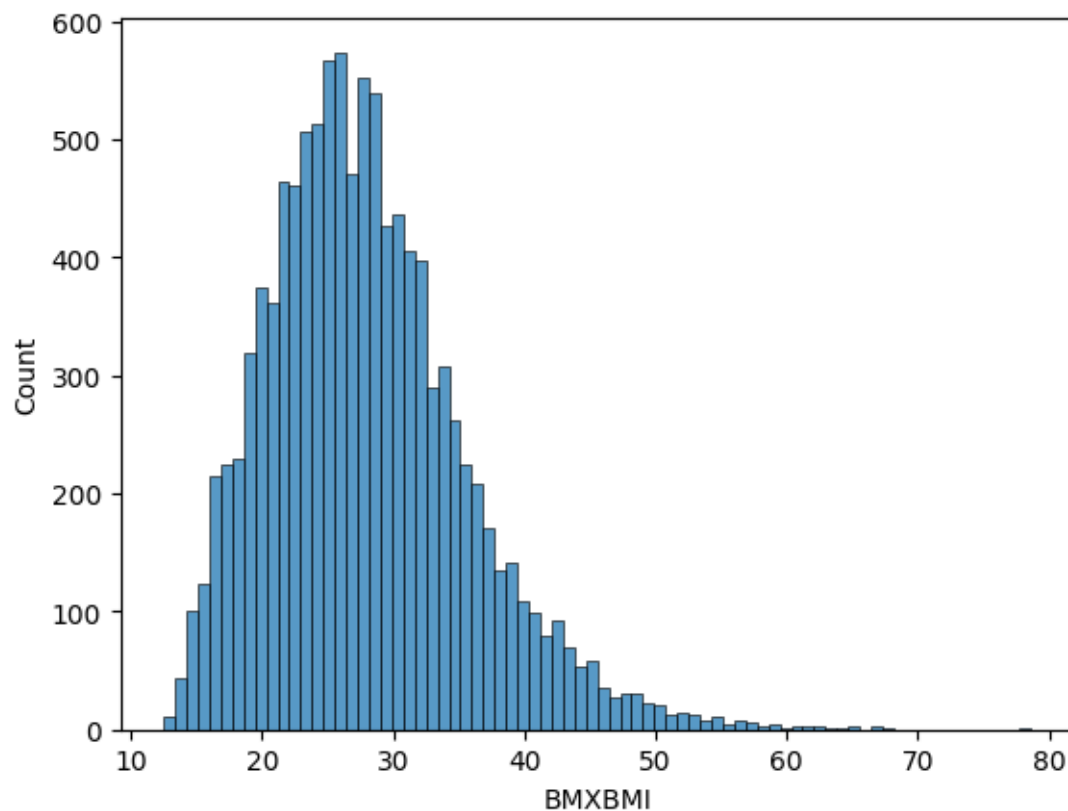
```
[119]: nhane.loc[nhane['BMXBMI']==nhane['BMXBMI'].max(), 'BMXBMI']
```

```
[119]: interview_day  
2018-09-12    92.3  
Name: BMXBMI, dtype: float64
```

```
[120]: nhane.loc[nhane['BMXBMI']==nhane['BMXBMI'].max(), 'BMXBMI'] = 30
```

```
[121]: tmp4 = nhane[['BMXBMI', 'BMXLEG', 'BMXWAIST', 'BMXHT']].copy()  
# tmp4.loc[tmp4['BMXBMI']==tmp4['BMXBMI'].max()]  
tmp4.dropna(inplace=True)  
sns.histplot(tmp4, x='BMXBMI')
```

```
[121]: <Axes: xlabel='BMXBMI', ylabel='Count'>
```



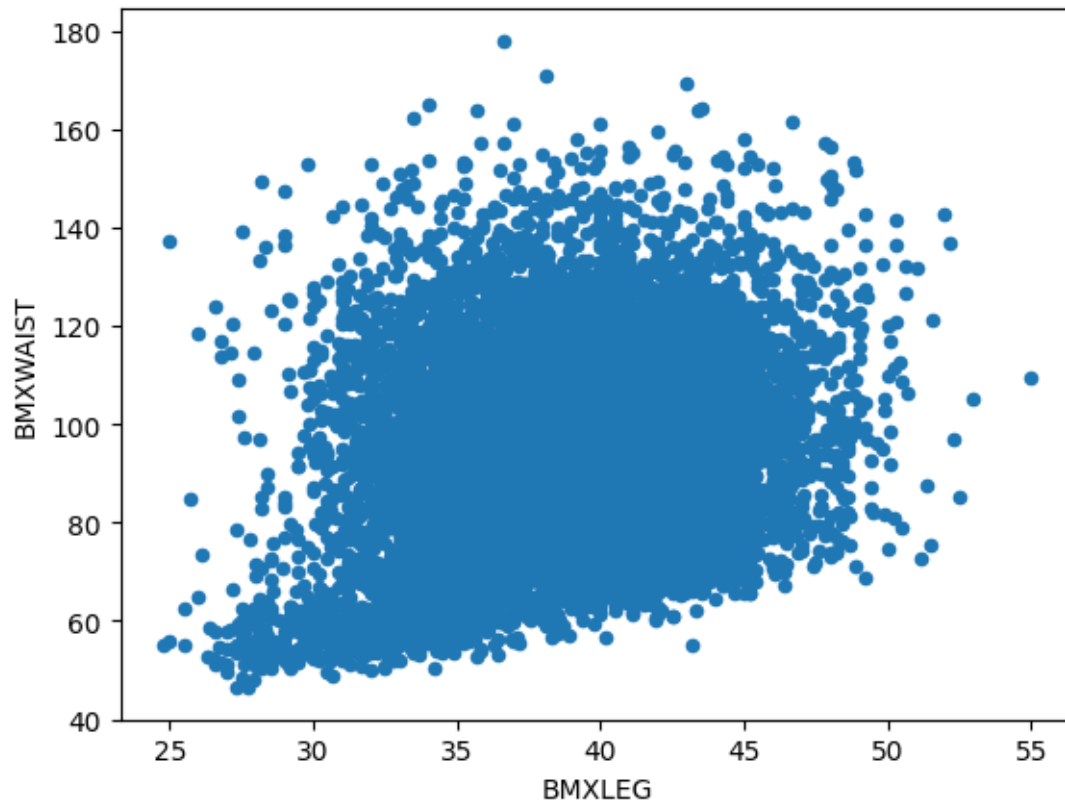
```
[122]: tmp4.loc[tmp4['BMXBMI']==tmp4['BMXBMI'].max()]
```

```
[122]:
```

	BMXBMI	BMXLEG	BMXWAIST	BMXHT
interview_day				
2017-10-16	78.6	35.3	153.0	159.9

```
[123]: nhane.plot(x='BMXLEG',y='BMXWAIST',kind='scatter')
```

```
[123]: <Axes: xlabel='BMXLEG', ylabel='BMXWAIST'>
```

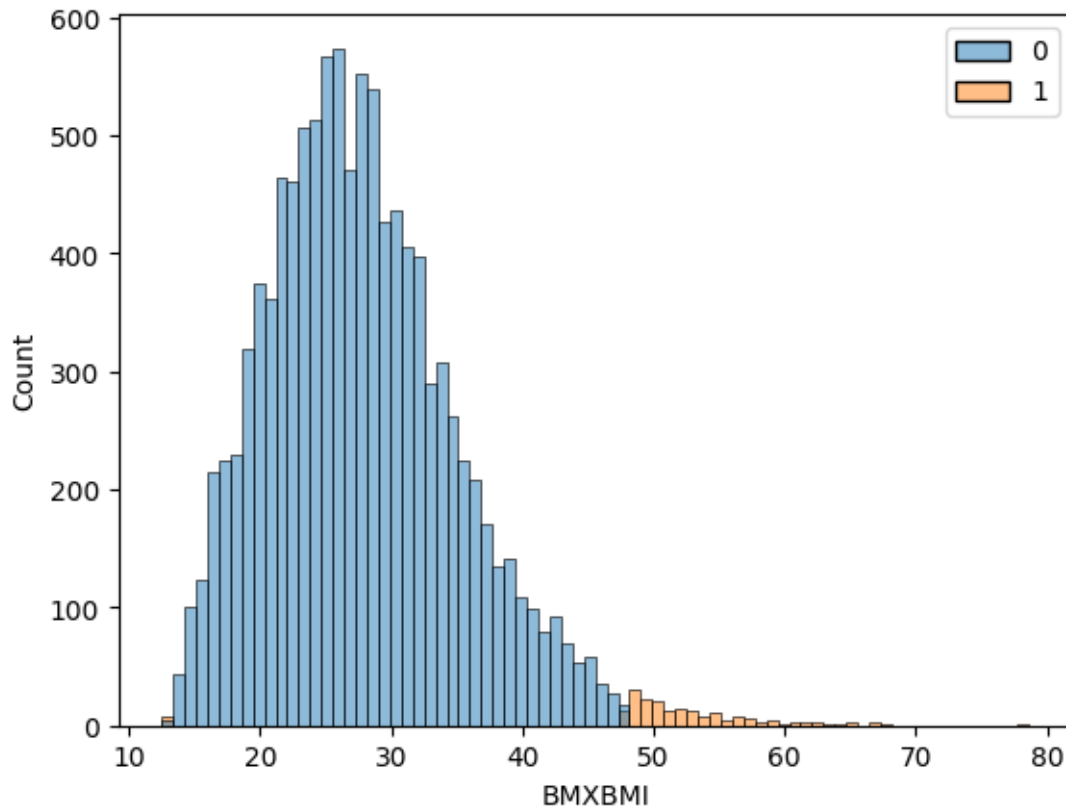


7.3.2 Use algorithm

```
[124]: from pyod.models.kde import KDE
      clf = KDE(contamination=0.018)
      clf.fit(tmp4['BMXBMI'].to_numpy().reshape(-1,1))
      y_train_pred = clf.labels_

      sns.histplot(tmp4,x='BMXBMI',hue=y_train_pred)
      np.unique(y_train_pred, return_counts=True)
```

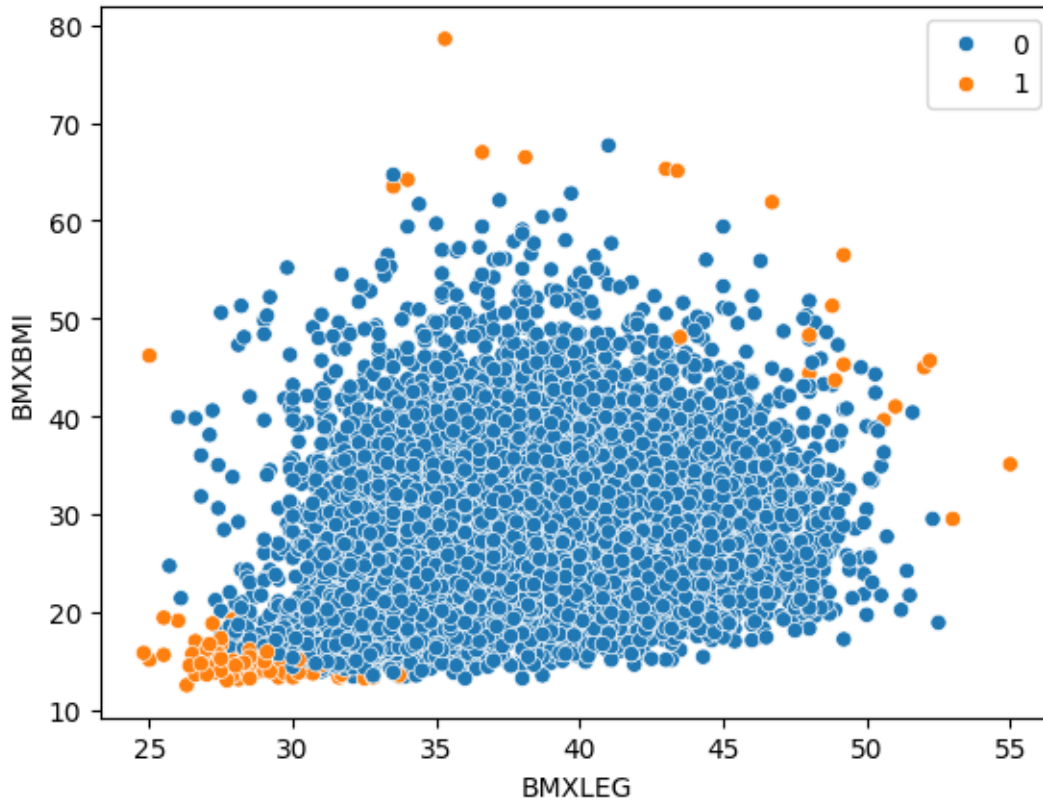
```
[124]: (array([0, 1]), array([10685,   196], dtype=int64))
```



```
[125]: from pyod.models.ecod import ECOD
from pyod.utils.data import evaluate_print
clf = ECOD(contamination=0.01)
clf.fit(tmp4)
y_train_pred = clf.labels_

sns.scatterplot(tmp4,x='BMXLEG',y='BMXBMI',hue=y_train_pred)
np.unique(y_train_pred, return_counts=True)
```

```
[125]: (array([0, 1]), array([10772, 109], dtype=int64))
```



```
[126]: nhane.reset_index(inplace=True)
```

Practice

- extract the rows with distinct date with complete survey status(keep the first observation)
- ['BMDSTATS'] 'Complete data for age group'
- use the algorithm above to detect outlier(BMXLEG)

```
[127]: nhane['BMDSTATS'].unique()
```

```
[127]: array(['Complete data for age group', 'Other partial exam', nan,
            'Partial: Only height and weight obtained',
            'No body measures exam data'], dtype=object)
```

```
[128]: nhane.drop_duplicates(subset=['interview_date'], keep='first', inplace=True)
```

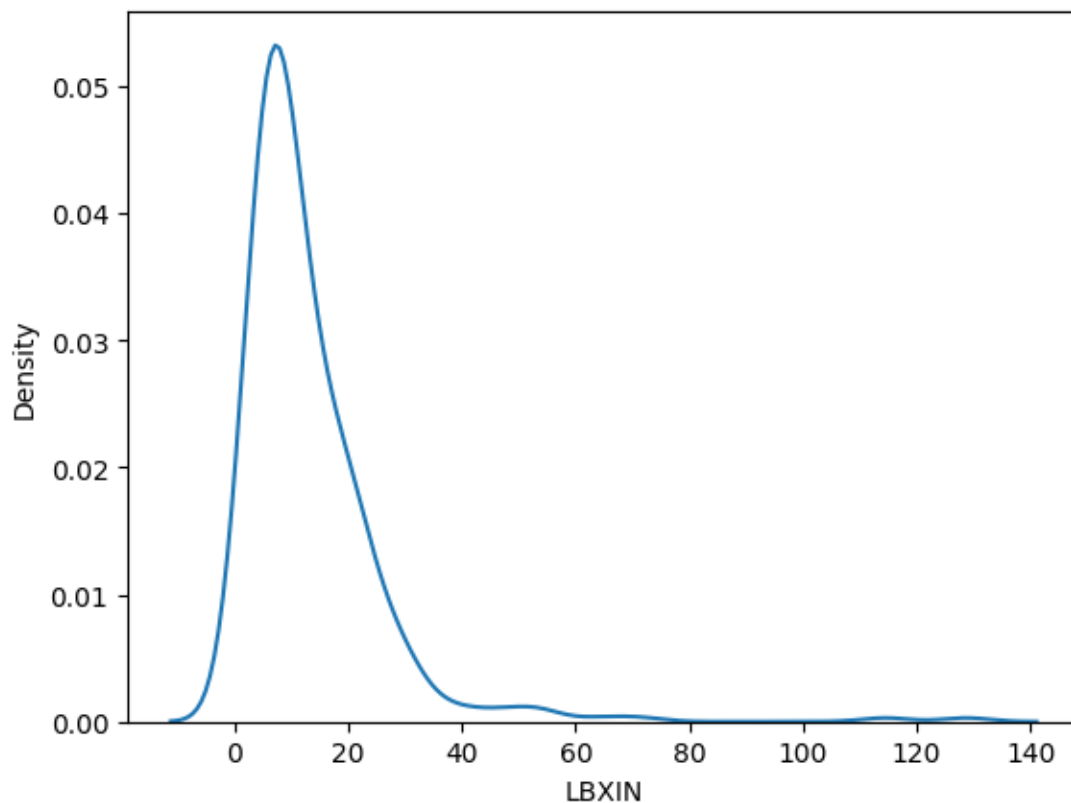
8 Imputation

1. by mean, median, mode
2. by interpolation linear regression, linear interpolation <https://pandas.pydata.org/pandas-docs/stable/reference/api/pandas.DataFrame.interpolate.html>

```
[129]: # nhane.loc[nhane['SEQN'] == 109513, 'LBXIN'] = 8
```

```
[130]: sns.kdeplot(nhane.reset_index(), x='LBXIN')
```

```
[130]: <Axes: xlabel='LBXIN', ylabel='Density'>
```



8.1 Impute by mean, median, mode

```
[131]: nhane['LBXIN'].fillna(nhane['LBXIN'].median())  
nhane['LBXIN'].fillna(nhane['LBXIN'].mean())  
# nhane['LBXIN'].fillna(nhane['LBXIN'].mode())
```

```
[131]: 0      13.374207  
16     13.374207  
32      4.720000  
45     13.374207  
60      7.670000  
...  
15493  13.374207  
15503  21.300000  
15520  13.374207
```

```

15533      5.540000
15545     13.374207
Name: LBXIN, Length: 1095, dtype: float64

```

8.2 Impute by interpolation

```
[132]: nhane.head()
```

```

[132]:   interview_day   SEQN BPQ080 BPQ100D      BMDSTATS  BMXHT  \
0      2017-01-01  112413     NaN     NaN  Complete data for age group  163.9
16     2017-01-02  123252     Yes     NaN  Complete data for age group  162.7
32     2017-01-03  115036     No     NaN  Complete data for age group  159.6
45     2017-01-04  116684     No     NaN  Complete data for age group  169.2
60     2017-01-05  121102     No     NaN  Complete data for age group  168.6

      BMXBMI      BMDMIC  BMXLEG  BMXWAIST  ...  LBXIN  MCQ053  LBXTC  \
0      15.6  Normal weight    39.2      65.4  ...   NaN     No   NaN
16     35.1           NaN    36.1     115.8  ...   NaN     No  208.0
32     23.6           NaN    36.8      80.5  ...  4.72     No  168.0
45     32.0           NaN    39.7     107.7  ...   NaN     No   NaN
60     28.9           NaN    39.0     101.2  ...  7.67     No  200.0

      interview_date  Day  Month  Year  Dayname  Day_week  daydiff
0      2017-01-01    1     1  2017   Sunday        6  -266 days
16     2017-01-02    2     1  2017   Monday        0   -2 days
32     2017-01-03    3     1  2017  Tuesday        1 -121 days
45     2017-01-04    4     1  2017 Wednesday        2 -667 days
60     2017-01-05    5     1  2017 Thursday        3 -565 days

```

```
[5 rows x 32 columns]
```

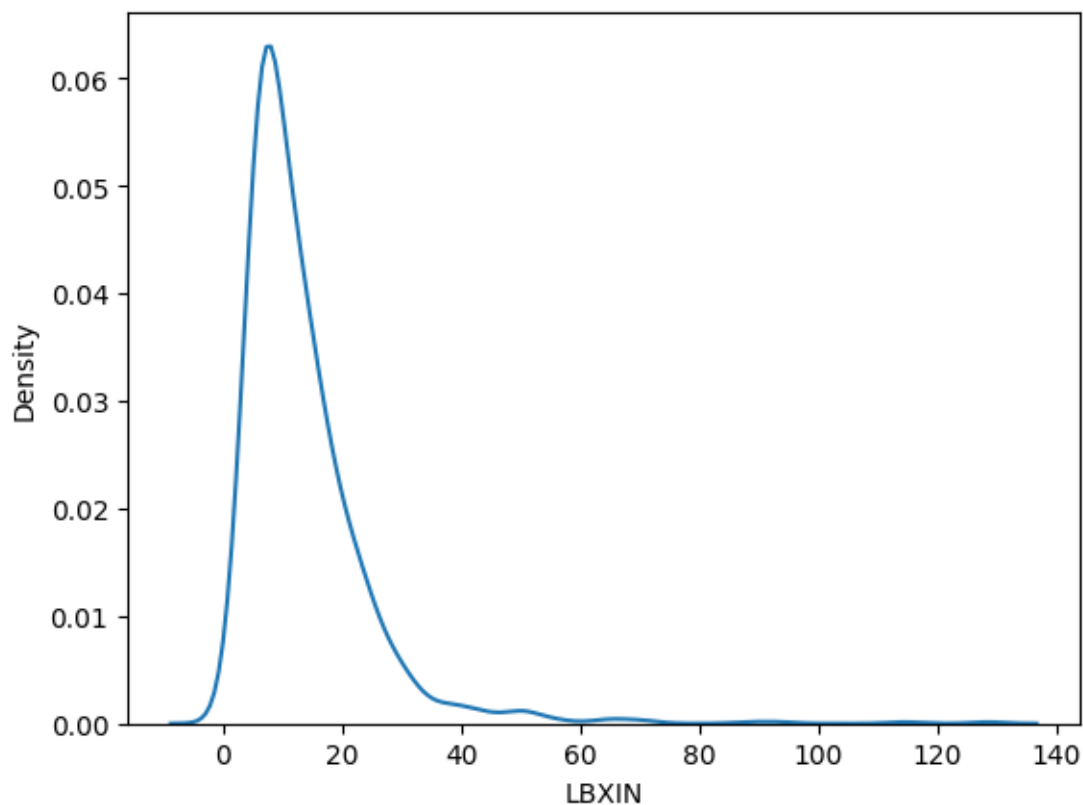
```

[133]: tmp5 = nhane['LBXIN'].copy()
tmp5.interpolate('spline', order=1, inplace=True)
tmp5.fillna(tmp5.mean(), inplace=True)

sns.kdeplot(tmp5.reset_index(), x='LBXIN')

```

```
[133]: <Axes: xlabel='LBXIN', ylabel='Density'>
```



```
[134]: tmp6 = nhane[['LBXIN', 'LBXTC', 'LUXCAPM']].copy()
      tmp6.interpolate(method='spline', order=1, axis=0)
```

```
[134]:
```

	LBXIN	LBXTC	LUXCAPM
0	NaN	NaN	NaN
16	NaN	208.000000	265.0
32	4.720000	168.000000	240.0
45	5.639628	182.857544	276.0
60	7.670000	200.000000	234.0
...	...	...	...
15493	22.431534	178.000000	220.0
15503	21.300000	159.000000	343.0
15520	12.371250	196.392243	208.0
15533	5.540000	225.000000	250.0
15545	-0.752082	153.000000	248.0

[1095 rows x 3 columns]

8.3 Practice

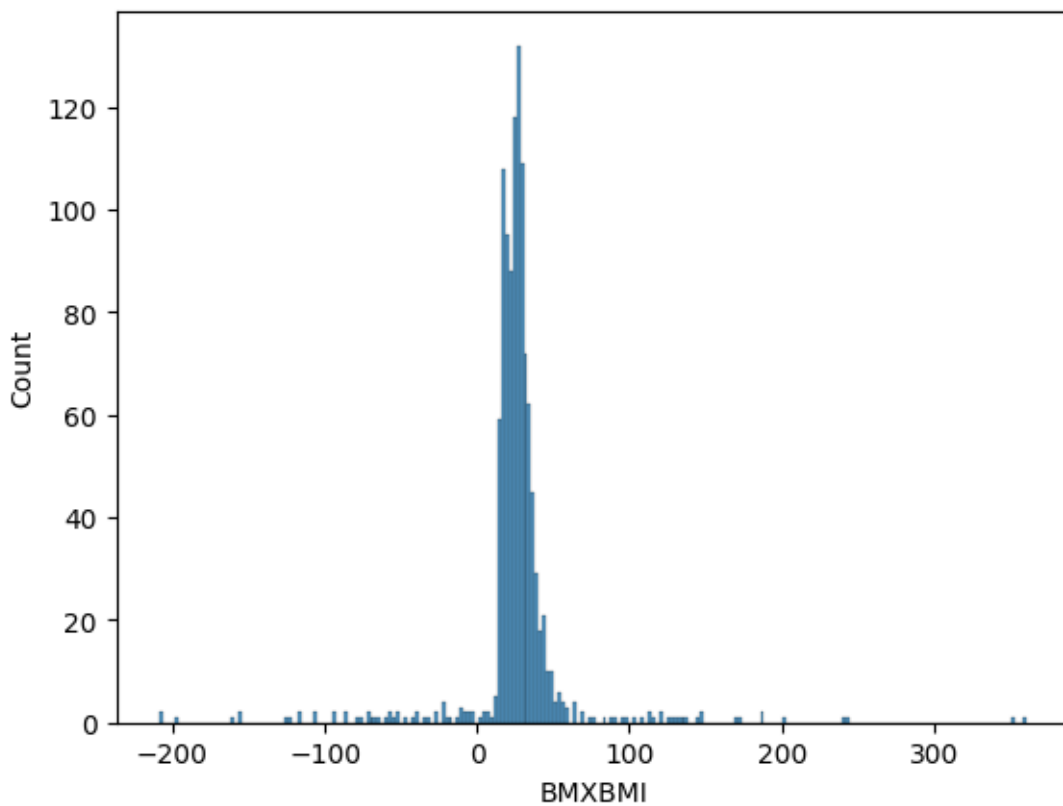
- interpolate fill nans in BMI with method above

```
[ ]:
```

```
[135]: tmp8 = nhane['BMXBMI'].copy()
tmp8.interpolate(method='spline', order=2, axis=0, inplace=True,
                  limit_area='inside', limit_direction='forward')
# tmp8.interpolate(method='spline', order=2,
#                  ↪axis=0, inplace=True, limit_direction='forward', downcast='integer')
# tmp8.fillna(tmp8.mean(), inplace=True)
```

```
[136]: sns.histplot(tmp8)
```

```
[136]: <Axes: xlabel='BMXBMI', ylabel='Count'>
```



9 Groupby method

```
[137]: nhane.groupby('BMDBMIC')[['LBXIN', 'LBXTC']].mean()
```

```
[137]:
```

	LBXIN	LBXTC
BMDBMIC		
Normal weight	9.402286	153.016667

Obese	26.677500	165.625000
Overweight	21.095000	158.575758
Underweight	7.855000	155.000000

```
[138]: nhane.groupby(['BMDBMIC', 'RIAGENDR'])[['LBXIN', 'LBXTC']].mean()
```

```
[138]:
```

		LBXIN	LBXTC
BMDBMIC	RIAGENDR		
Normal weight	Female	9.636111	152.920635
	Male	9.154706	153.122807
Obese	Female	25.092500	168.208333
	Male	28.262500	163.041667
Overweight	Female	12.993333	161.909091
	Male	29.196667	151.909091
Underweight	Female	7.855000	156.250000
	Male	NaN	150.000000

9.1 Practice

- calculate group BMI by race
- calculate group BMI by race and gender

```
[ ]:
```

```
[ ]:
```

10 Crosstab method

```
[139]: pd.crosstab(nhane['BMDBMIC'], nhane['RIAGENDR'])
```

```
[139]:
```

	Female	Male
BMDBMIC		
Normal weight	102	96
Obese	34	37
Overweight	32	26
Underweight	8	4

```
[140]: pd.crosstab(nhane['BMDBMIC'], nhane['RIAGENDR'], values=nhane['LBXIN'],
↪, aggfunc='mean')
```

```
[140]:
```

	Female	Male
BMDBMIC		
Normal weight	9.636111	9.154706
Obese	25.092500	28.262500
Overweight	12.993333	29.196667
Underweight	7.855000	NaN

10.1 Practice

- calculate group BMI by race use crosstab method
- calculate group BMI by race and gender use crosstab method

```
[ ]:
```

```
[ ]:
```

11 Data Transpose

1. long to wide
2. wide to long

11.1 long to wide

```
[141]: tmp7 = nhane[['RIAGENDR', 'LBXIN', 'BMDBMIC']].copy()
# tmp7.reset_index(inplace=True)
# tmp7.drop(columns='interview_date', axis=1, inplace=True)
tmp7.dropna(inplace=True)
tmp7.sort_values(['RIAGENDR', 'BMDBMIC'], inplace=True)
```

```
[142]: tmp7.head()
```

```
[142]:
```

	RIAGENDR	LBXIN	BMDBMIC
288	Female	27.37	Normal weight
405	Female	4.04	Normal weight
817	Female	6.44	Normal weight
1532	Female	11.22	Normal weight
1926	Female	7.72	Normal weight

```
[143]: tmp7['ids'] = tmp7.groupby(['RIAGENDR', 'BMDBMIC']).cumcount() # add unique id_
↳ in each row to avoid duplicate index problem
```

```
[144]: tmp7.pivot(index=['RIAGENDR', 'ids'], columns='BMDBMIC', values='LBXIN')
```

```
[144]:
```

BMDBMIC		Normal weight	Obese	Overweight	Underweight
RIAGENDR	ids				
Female	0	27.37	39.26	7.60	10.56
	1	4.04	19.12	7.23	5.15
	2	6.44	23.30	24.15	NaN
	3	11.22	18.69	NaN	NaN
	4	7.72	NaN	NaN	NaN
	5	5.04	NaN	NaN	NaN
	6	6.98	NaN	NaN	NaN
	7	14.68	NaN	NaN	NaN
	8	10.55	NaN	NaN	NaN
	9	6.59	NaN	NaN	NaN

	10	14.49	NaN	NaN	NaN
	11	13.36	NaN	NaN	NaN
	12	17.15	NaN	NaN	NaN
	13	7.61	NaN	NaN	NaN
	14	4.20	NaN	NaN	NaN
	15	3.73	NaN	NaN	NaN
	16	6.14	NaN	NaN	NaN
	17	6.14	NaN	NaN	NaN
Male	0	11.69	12.32	12.32	NaN
	1	6.21	21.45	25.84	NaN
	2	10.76	51.59	49.43	NaN
	3	4.33	27.69	NaN	NaN
	4	3.42	NaN	NaN	NaN
	5	9.08	NaN	NaN	NaN
	6	7.35	NaN	NaN	NaN
	7	6.79	NaN	NaN	NaN
	8	20.70	NaN	NaN	NaN
	9	9.51	NaN	NaN	NaN
	10	15.85	NaN	NaN	NaN
	11	6.70	NaN	NaN	NaN
	12	14.13	NaN	NaN	NaN
	13	6.26	NaN	NaN	NaN
	14	1.75	NaN	NaN	NaN
	15	10.17	NaN	NaN	NaN
	16	10.93	NaN	NaN	NaN

```
[145]: tmp7.pivot(index=['BMDBMIC', 'ids'], columns='RIAGENDR', values='LBXIN')
```

```
[145]: RIAGENDR      Female  Male
BMDBMIC  ids
Normal weight 0      27.37  11.69
              1       4.04   6.21
              2       6.44  10.76
              3      11.22   4.33
              4       7.72   3.42
              5       5.04   9.08
              6       6.98   7.35
              7      14.68   6.79
              8      10.55  20.70
              9       6.59   9.51
             10      14.49  15.85
             11      13.36   6.70
             12      17.15  14.13
             13       7.61   6.26
             14       4.20   1.75
             15       3.73  10.17
             16       6.14  10.93
```

	17	6.14	NaN
Obese	0	39.26	12.32
	1	19.12	21.45
	2	23.30	51.59
	3	18.69	27.69
Overweight	0	7.60	12.32
	1	7.23	25.84
	2	24.15	49.43
Underweight	0	10.56	NaN
	1	5.15	NaN

11.2 wide to long

```
[146]: tmp8 = tmp7.pivot(index=['BMDBMIC', 'ids'], columns='RIAGENDR', values='LBXIN')
      tmp8
```

```
[146]: RIAGENDR      Female   Male
BMDBMIC  ids
Normal weight 0      27.37  11.69
              1       4.04   6.21
              2       6.44  10.76
              3      11.22   4.33
              4       7.72   3.42
              5       5.04   9.08
              6       6.98   7.35
              7      14.68   6.79
              8      10.55  20.70
              9       6.59   9.51
             10      14.49  15.85
             11      13.36   6.70
             12      17.15  14.13
             13       7.61   6.26
             14       4.20   1.75
             15       3.73  10.17
             16       6.14  10.93
             17       6.14   NaN
Obese      0      39.26  12.32
           1      19.12  21.45
           2      23.30  51.59
           3      18.69  27.69
Overweight 0       7.60  12.32
           1       7.23  25.84
           2      24.15  49.43
Underweight 0      10.56   NaN
           1       5.15   NaN
```

```
[147]: df_wide_reset = tmp8.reset_index()
df_wide_reset.head()
```

```
[147]: RIAGENDR      BMDBMIC  ids  Female  Male
0      Normal weight    0    27.37  11.69
1      Normal weight    1     4.04   6.21
2      Normal weight    2     6.44  10.76
3      Normal weight    3    11.22   4.33
4      Normal weight    4     7.72   3.42
```

```
[148]: # Melting the wide-format dataframe back to long-format
df_long_resaped = pd.melt(df_wide_reset, id_vars=['BMDBMIC', 'ids'],
    value_vars=['Female', 'Male'],
    var_name='Gender', value_name='LBXIN')
df_long_resaped
```

```
[148]:      BMDBMIC  ids  Gender  LBXIN
0  Normal weight    0  Female  27.37
1  Normal weight    1  Female   4.04
2  Normal weight    2  Female   6.44
3  Normal weight    3  Female  11.22
4  Normal weight    4  Female   7.72
5  Normal weight    5  Female   5.04
6  Normal weight    6  Female   6.98
7  Normal weight    7  Female  14.68
8  Normal weight    8  Female  10.55
9  Normal weight    9  Female   6.59
10 Normal weight   10  Female  14.49
11 Normal weight   11  Female  13.36
12 Normal weight   12  Female  17.15
13 Normal weight   13  Female   7.61
14 Normal weight   14  Female   4.20
15 Normal weight   15  Female   3.73
16 Normal weight   16  Female   6.14
17 Normal weight   17  Female   6.14
18      Obese      0  Female  39.26
19      Obese      1  Female  19.12
20      Obese      2  Female  23.30
21      Obese      3  Female  18.69
22  Overweight    0  Female   7.60
23  Overweight    1  Female   7.23
24  Overweight    2  Female  24.15
25  Underweight    0  Female  10.56
26  Underweight    1  Female   5.15
27  Normal weight    0    Male  11.69
28  Normal weight    1    Male   6.21
29  Normal weight    2    Male  10.76
```

30	Normal weight	3	Male	4.33
31	Normal weight	4	Male	3.42
32	Normal weight	5	Male	9.08
33	Normal weight	6	Male	7.35
34	Normal weight	7	Male	6.79
35	Normal weight	8	Male	20.70
36	Normal weight	9	Male	9.51
37	Normal weight	10	Male	15.85
38	Normal weight	11	Male	6.70
39	Normal weight	12	Male	14.13
40	Normal weight	13	Male	6.26
41	Normal weight	14	Male	1.75
42	Normal weight	15	Male	10.17
43	Normal weight	16	Male	10.93
44	Normal weight	17	Male	NaN
45	Obese	0	Male	12.32
46	Obese	1	Male	21.45
47	Obese	2	Male	51.59
48	Obese	3	Male	27.69
49	Overweight	0	Male	12.32
50	Overweight	1	Male	25.84
51	Overweight	2	Male	49.43
52	Underweight	0	Male	NaN
53	Underweight	1	Male	NaN

12 Apply() Method

For a Series: It applies a function element-wise to the values of the Series.

For a DataFrame: It applies the function along an axis (either rows or columns).

```
[149]: def cat(x):
        if (x>=0) & (x<20):
            return 'normal'
        elif (x>=20) & (x<30):
            return 'over weight'
        elif x is None:
            return 'no info'

        def calc(x):
            if x is None:
                return np.nan
            else:
                return x**2+6-2*x
```

```
[150]: nhane['BMXBMI'].apply(calc)
```

```
[150]: 0          218.16
      16         1167.81
      32          515.76
      45          966.00
      60          783.41
      ...
      15493       875.25
      15503      1441.41
      15520        630.00
      15533        696.69
      15545        511.25
      Name: BMXBMI, Length: 1095, dtype: float64
```

```
[151]: nhane['BMXBMI'].apply(cat)
```

```
[151]: 0          normal
      16           None
      32      over weight
      45           None
      60      over weight
      ...
      15493          None
      15503          None
      15520      over weight
      15533      over weight
      15545      over weight
      Name: BMXBMI, Length: 1095, dtype: object
```

12.1 Practice

- create a function called conversions. the body of the function is to normalize BMI columns as follows

$$\frac{(BMI - average(BMI))}{max(BMI) - min(BMI)}$$

```
[ ]:
```

```
[ ]:
```

13 Combine multiple dataset

- <https://statisticsglobe.com/combine-pandas-dataframes-vertically-horizontally-python>

13.1 Combine vertically

```
[152]: table1 = pd.DataFrame({"ID":range(1, 5),          # Create first pandas DataFrame
                             "x1":range(10, 6, - 1),
                             "x2":["a", "b", "c", "d"],
                             "x3":range(26, 22, - 1)})
```

table1

```
[152]:
```

	ID	x1	x2	x3
0	1	10	a	26
1	2	9	b	25
2	3	8	c	24
3	4	7	d	23

```
[153]: table2 = pd.DataFrame({"ID":range(2, 8),          # Create second pandas DataFrame
                             "x2":["x", "y", "y", "x", "y", "y"],
                             "x1":range(12, 1, - 2)})
```

table2

```
[153]:
```

	ID	x2	x1
0	2	x	12
1	3	y	10
2	4	y	8
3	5	x	6
4	6	y	4
5	7	y	2

append table1 (x1,x2) above table2 (y1,y2)

```
[154]: pd.concat([table1[["x1", "x2"]], table2[["x2", "x1"]]],  
                ↪axis=0,ignore_index=True)
```

```
[154]:
```

	x1	x2
0	10	a
1	9	b
2	8	c
3	7	d
4	12	x
5	10	y
6	8	y
7	6	x
8	4	y
9	2	y

13.2 Combine horizontally (by a common column)

```
[155]: table1.merge(table2, on="ID", how="inner")
```

```
[155]:
```

	ID	x1_x	x2_x	x3	x2_y	x1_y
0	2	9	b	25	x	12
1	3	8	c	24	y	10
2	4	7	d	23	y	8

```
[156]: table1.merge(table2, on="ID", how="left")
```

```
[156]:
```

	ID	x1_x	x2_x	x3	x2_y	x1_y
0	1	10	a	26	NaN	NaN
1	2	9	b	25	x	12.0
2	3	8	c	24	y	10.0
3	4	7	d	23	y	8.0

```
[157]: table1.merge(table2, on="ID", how="right")
```

```
[157]:
```

	ID	x1_x	x2_x	x3	x2_y	x1_y
0	2	9.0	b	25.0	x	12
1	3	8.0	c	24.0	y	10
2	4	7.0	d	23.0	y	8
3	5	NaN	NaN	NaN	x	6
4	6	NaN	NaN	NaN	y	4
5	7	NaN	NaN	NaN	y	2

```
[158]: table1.merge(table2, on="ID", how="outer")
```

```
[158]:
```

	ID	x1_x	x2_x	x3	x2_y	x1_y
0	1	10.0	a	26.0	NaN	NaN
1	2	9.0	b	25.0	x	12.0
2	3	8.0	c	24.0	y	10.0
3	4	7.0	d	23.0	y	8.0
4	5	NaN	NaN	NaN	x	6.0
5	6	NaN	NaN	NaN	y	4.0
6	7	NaN	NaN	NaN	y	2.0

Practice

- Use file I provided to appropriately combine the datasets into one dataset.

14 time series data

```
[159]: import pandas as pd
df = pd.read_csv('heartrate_seconds_merged.csv')
df['Time'] = pd.to_datetime(df['Time'])
# df.set_index('Time', inplace=True)
```

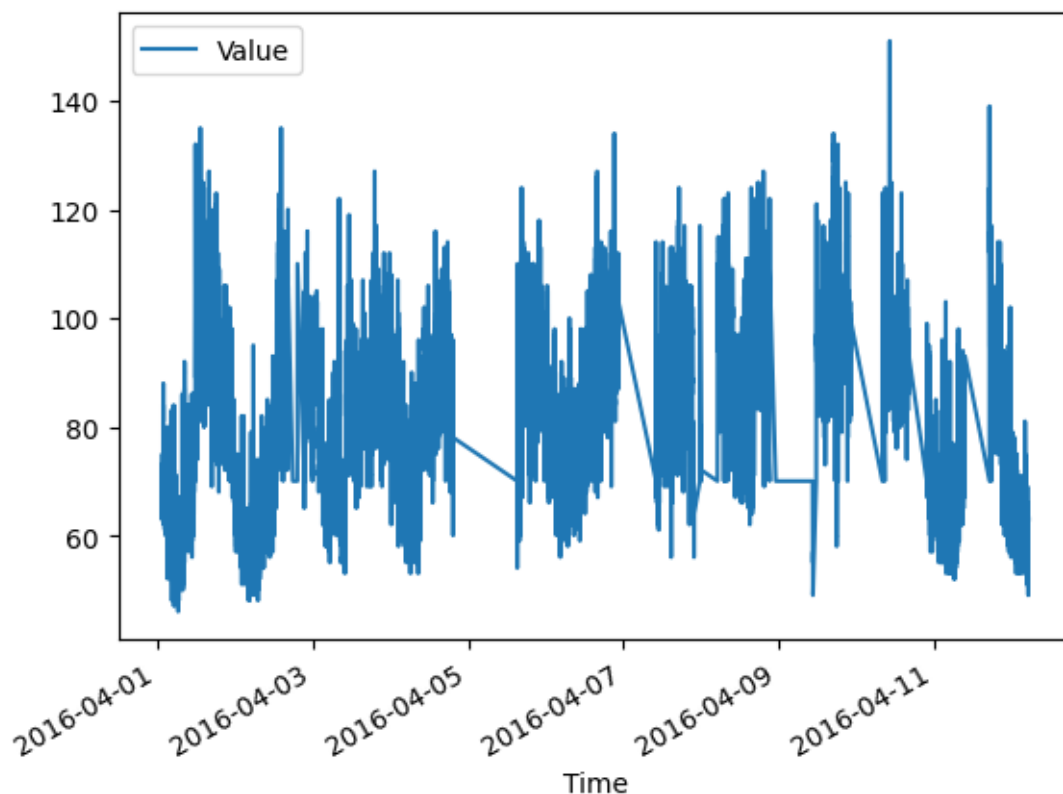
```
df = df[df['Id'] == 4020332650].copy()
df.sort_values(by='Time', inplace=True, ascending=True)
df.drop(['Id'], axis=1, inplace=True)
df.head()
```

```
[159]:
```

		Time	Value
177726	2016-04-01	01:03:18	71
177727	2016-04-01	01:03:21	71
177728	2016-04-01	01:03:24	71
177729	2016-04-01	01:03:27	71
177730	2016-04-01	01:03:30	71

```
[160]: df.plot(x='Time', y='Value')
```

```
[160]: <Axes: xlabel='Time'>
```



```
[161]: df['daydiff'] = df['Time'].diff()
```

```
[162]: df['daydiff'].dt.components['hours'].unique()
```

```
[162]: array([nan,  0.,  1.,  2., 19., 11.,  4.,  9.,  5.,  7.])
```

```
[163]: df[df['daydiff'].dt.components['hours']>=2]
```

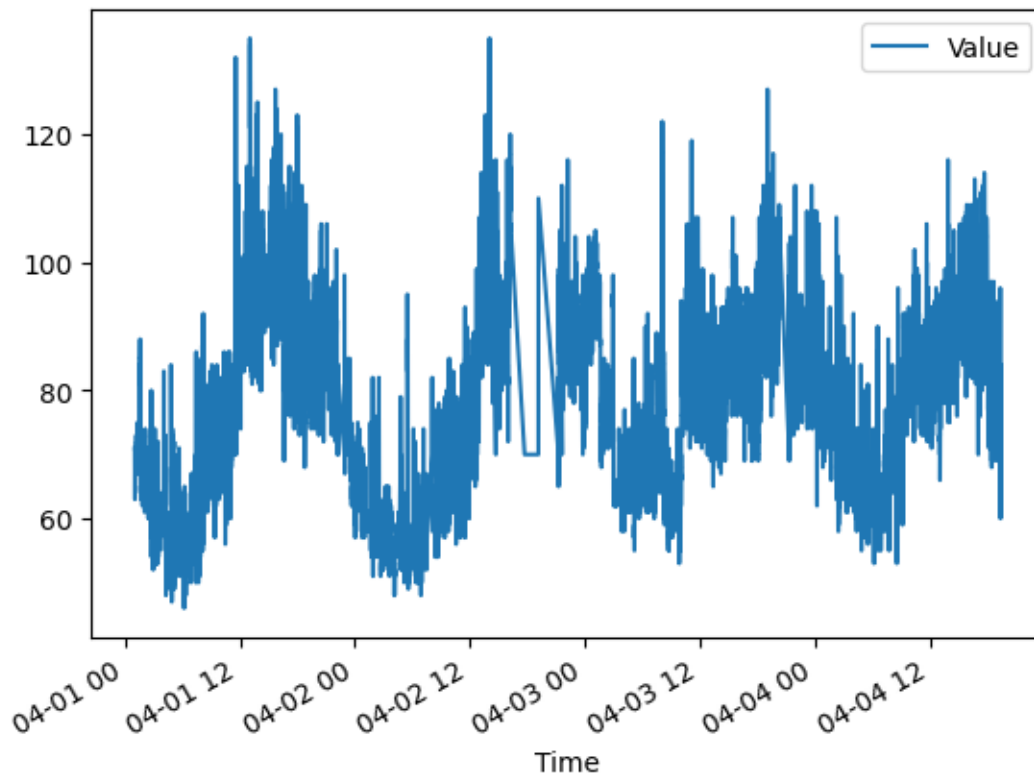
```
[163]:
```

		Time	Value	daydiff
236116	2016-04-02	21:12:23	70	0 days 02:05:19
304768	2016-04-05	15:06:03	70	0 days 19:47:05
352808	2016-04-07	10:03:31	70	0 days 11:37:02
370840	2016-04-08	05:02:52	70	0 days 04:56:08
396787	2016-04-09	10:28:46	70	0 days 11:23:58
413258	2016-04-10	08:03:57	70	0 days 09:53:28
424551	2016-04-10	21:49:17	70	0 days 05:45:20
442887	2016-04-11	17:13:15	70	0 days 07:34:17

```
[164]: df.set_index('Time',inplace=True)
```

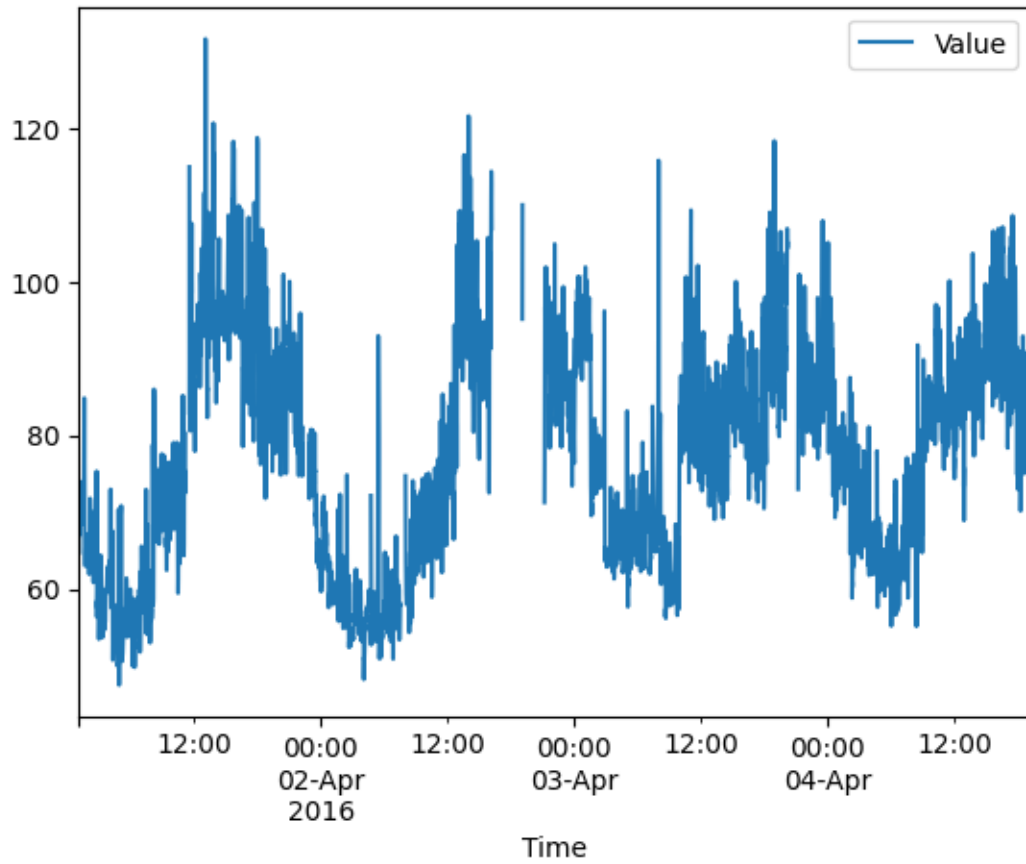
```
[165]: df1 = df.loc[:'2016-04-04'].copy()  
df1.drop('daydiff',axis=1,inplace=True)  
df1.plot()
```

```
[165]: <Axes: xlabel='Time'>
```



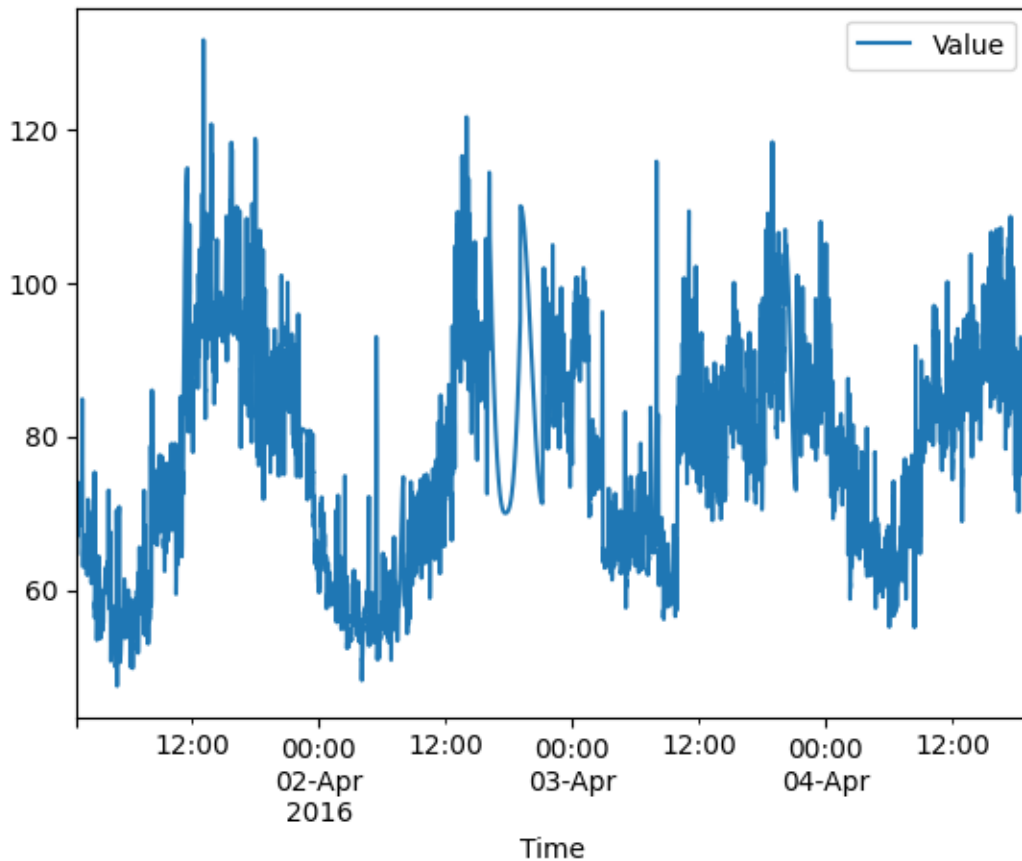
```
[166]: df2 = df1.resample('1min').mean()  
df2.plot()
```

```
[166]: <Axes: xlabel='Time'>
```



```
[167]: df2.interpolate(method='pchip').plot()
```

```
[167]: <Axes: xlabel='Time'>
```



```
[168]: df2.interpolate(method='pchip',inplace=True)
```

15 train - test split - None time series

```
[169]: import pandas as pd
from sklearn.model_selection import train_test_split

feature = [x for x in nhane.columns if x != 'LBXIN']

X = nhane[feature] # Features
y = nhane['LBXIN'] # Target

# Perform train-test split (80% train, 20% test)
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2,
↳ random_state=42)
```

train - test split with no labels

```
[170]: X_train, X_test = train_test_split(X, test_size=0.2, random_state=42)
```

```
[171]: df2.to_numpy()
```

```
[171]: array([[68.68421053],  
            [64.51851852],  
            [68.42307692],  
            ...,  
            [77.4137931 ],  
            [79.44      ],  
            [80.37037037]])
```

15.0.1 heart dataset into train and test set - Using sktime.

```
[172]: from sktime.split import temporal_train_test_split  
X_train, X_test = temporal_train_test_split(df2, test_size=0.2)
```

15.1 Homework

1. Look at the nhanes dataset as a whole, select appropriate observations as your samples and state the reason use python code to demonstrate your result.
2. Based on the steps above, detect outlier, delete duplicate values, and fill NaNs use methods above.

```
[ ]:
```